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Leading an emerging technology towards purposeful transitions

A socio-technical exploration of civilian drones

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Abstract

Rapid technological change is driving complex sustainability transitions in unpredictable ways, bringing both opportunities and systemic uncertainty. One such technology are civilian drones, emerging as a mode of transport within broader agendas for resilient, electrified, and sustainable transport which promises to improve accessibility, congestion, and services. Its growing technological readiness and evolving regulatory frameworks brings forth complex systems dynamics to transport transition which largely remains unexplored. This thesis develops a foundational understanding of drone transport as an emerging socio-technical system within broader sustainable transport transitions. This research focuses on its impacts, systems interactions, governance challenges, and implications for low-carbon and energy-efficient mobility. A combination of methods such as literature reviews, semi-structured interviews, narrative analysis, energy and emission assessments, life-cycle costs, are employed along with a co-creative approach to explore this new field. Potential barriers and drivers for drone transport implementation are identified in the thesis based on their impacts and systems interactions. Results show how its impacts, particularly for last-mile goods delivery, are strongly influenced by the type of drone and how it is implemented such as the spatial context, operational parameters, and what vehicles are being complemented or substituted. Further, regulatory uncertainty, competing discourses, and complex planning challenges that interact with public acceptance are revealed and addressed to align their development towards local transport needs. Moreover, results provide insights for drones and other emerging technologies to be understood critically, with nuanced discourses on technology, coordinated policy and planning approaches, engagements in transparent dialogue, to actively steer their development in early stages toward purposeful transitions.

Keywords: civilian drones, Urban Air Mobility, Advanced Air Mobility, Innovative Air Mobility and Services, emerging technology, sustainability transitions, multi-level perspective, electric vehicle, last-mile logistics, multimodal transport

Sammanfattning

Snabb teknisk förändring driver komplexa hållbarhetsomställningar på oförutsägbara sätt och medför både möjligheter och systemisk osäkerhet. En sådan teknik är civila drönare, som växer fram som ett transportsätt för robust, elektrifierad och hållbar transport och som utlovar förbättrad tillgänglighet, minskad trängsel och förbättrade tjänster. Dess ökande tekniska mognad och utvecklande regelverk ger upphov till komplex systemdynamik i transportomställningen som till stor del fortfarande är outforskad. Denna avhandling utvecklar en grundläggande förståelse av drönartransport som ett framväxande socio-tekniskt system inom bredare hållbara transportomställningar. Forskningen fokuserar på dess effekter, systeminteraktioner, styrningsutmaningar samt implikationer för koldioxidsnål och energieffektiv mobilitet. En kombination av metoder, såsom litteraturstudier, semistrukturerade intervjuer, narrativ analys samt energi- och emissionsbedömningar och livscykelkostnadsberäkningar, används tillsammans med ett samskapande angreppssätt för att utforska detta nya fält. Potentiella hinder och drivkrafter för implementering av drönartransport identifieras i avhandlingen utifrån deras effekter och systeminteraktioner. Resultaten visar hur effekterna, särskilt för leveranser av gods i sista milen, i hög grad påverkas av typen av drönare och hur den implementeras, såsom den rumsliga kontexten, operativa parametrar och vilka fordon som kompletteras eller ersätts. Vidare synliggörs och behandlas regulatorisk osäkerhet, konkurrerande diskurser och komplexa planeringsutmaningar som samverkar med allmän acceptans, i syfte att rikta utvecklingen mot lokala transportbehov. Slutligen ger resultaten insikter för hur drönare och andra framväxande teknologier kan förstås kritiskt, genom nyanserade teknikdiskurser, samordnade policy- och planeringsansatser samt engagemang i transparent dialog, för att aktivt styra deras utveckling i tidiga skeden mot målmedvetna omställningar.

Nyckelord: civila drönare, Urban Air Mobility, Advanced Air Mobility, Innovative Air Mobility and Services, framväxande teknik, hållbarhetsomställning, multi-level perspektiv, elfordon, logistik för sista milen, multimodala transporter

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List of Papers

This thesis is based on the following papers, which are referred to in the text by Roman numerals.

Paper I
Kulanovic, A., Raghunatha, A., Nordensvärd, J., & Thollander, P. (2025). Analyzing discursive policy leadership using regime narratives in Sweden's emerging drone transport for sustainability transition. Sustainable Futures, 10, 101387. https://doi.org/10.1016/j.sftr.2025.101387

Paper II
Raghunatha, A., Thollander, P., & Barthel, S. (2023). Addressing the emergence of drones–A policy development framework for regional drone transportation systems. Transportation Research Interdisciplinary Perspectives, 18, 100795. https://doi.org/10.1016/j.trip.2023.100795

Paper III
Raghunatha, A., Hilletofth, P., & Thollander, P. (2025). Exploring the dynamics of public acceptance factors and drone transport planning towards sustainable transitions. European Transport Research Review, 17(1), 53. https://doi.org/10.1186/s12544-025-00748-3

Paper IV
Raghunatha, A., Lindkvist, E., Thollander, P., Hansson, E., & Jonsson, G. (2023). Critical assessment of emissions, costs, and time for last-mile goods delivery by drones versus trucks. Scientific Reports, 13(1), 11814. https://doi.org/10.1038/s41598-023-38922-z

Paper V
Raghunatha, A., Hilletofth, P., Hillman, K., Thollander, P. Barthel, S., Satish, G. S (Submitted). The complementary role of drones in energy-efficient transport transitions

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Author contributions

For Paper I, Aishwarya Raghunatha contributed to data collection, analysis and co-authoring the original draft along with the lead author Aneta Kulanovic. Johan Nordensvard and Patrik Thollander supervised, contributed to the analysis, and co-authored the paper.

In Paper II, Aishwarya Raghunatha contributed and served as the lead author of the original draft, conducted the data collection, analysis, and validation. Patrik Thollander and Stephan Barthel supervised and co-authored the paper.

Aishwarya Raghunatha contributed to Paper III by data collection, analysis, validation, and led the writing of the original draft. Per Hilletofth and Patrik Thollander supervised and co-authored the paper.

In Paper IV, Aishwarya Raghunatha conceptualized the study for a master's thesis and supervised the thesis. Erika Hansson and Greta Jonsson carried out the data collection, methodology, calculations, analysis, and writing of the thesis report. Aishwarya Raghunatha led the writing of the original draft and analysis. Emma Lindkvist supervised, guided the methodology, and co-authored the paper. Patrik Thollander supervised and co-authored the paper.

In Paper V, Aishwarya Raghunatha conceptualized the study, developed the method, conducted calculations, analysis, and led the writing of the original draft. Per Hilletofth, Karl Hillman, and Patrik Thollander supported in conceptualization, supervised and co-authored the paper. Stephan Barthel supervised and co-authored the paper. Sajjan Gundmi Satish supported in the development of method and the MATLAB model.

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1. Introduction

We live in an era of accelerating and overlapping transitions, where the proliferation of emerging technologies drives complex sustainability transitions in unpredictable ways (Paredis, 2011) bringing both opportunities and systemic uncertainties. One such technology is the drone, or "unmanned aerial vehicle" (UAV) (alternatively unmanned aerial systems (UAS)), which is an airborne vehicle that is operated remotely with varying levels of autonomy without an onboard pilot. Invented in the late 20th and 21st centuries primarily for military purposes (Cohen et al., 2021). It has been further developed for on-demand civilian applications, primarily goods transport and emergency applications, with further developments toward passenger transport. This has been possible due to technological developments in electric propulsion, lightweight materials, battery technology, software, artificial intelligence, automation, and many such landscape innovations. Incidentally, in the mid and late 20th century, helicopters, with pilots onboard, were used for on-demand passenger and goods transport, which can be construed as the emergence of the concept of urban air mobility (Cohen et al., 2021). Regular usage of this has since been discontinued or used only for emergency purposes due to technical challenges and a lack of safety. Drones have since emerged as the primary and radical driver of the concept of urban air mobility, which received attention in the early 21st century, often with the intention to reduce congestion, air pollution, and faster deliveries (Holden & Goel, 2016). The concept has further evolved to applications in rural areas for providing better transport and mobility services (Park et al., 2018), construing the overarching term Advanced Air Mobility, still used in the USA. In Europe, the concept is referred to as 'Innovative air mobility' and 'innovative air services', which distinguishes drones for passenger mobility and delivery (including emergency services) applications, respectively (EASA, 2023).

During COVID, using drones as a mode of transport received particular attention for delivering medical commodities, delivering food supplies, etc to communities for essential services in times of reduced physical contact (Schaarschmidt et al., 2021; UNICEF, 2020). This has, since the early 2020s, led to further interest in the technology and its civilian use cases for increased resilience in transport. As an electric mode of transport, drones have received further attention to be explored in the context of the energy and transport transition (Garrow et al., 2021). EU drone strategy 2.0 reflects these intentions by developing pathways to leverage the potential of drones to impact sustainable mobility plans (EU, 2022b).

This has also brought forth air-taxi developments, or passenger drones with high payload (e.g., developments by Ehang, Lilium, etc). However, the higher technical capabilities for safe and feasible passenger drone transport in the built environment are some of the main challenges that are expected to have a step-

by-step development (Ahmed et al., 2020; Winter et al., 2020). Delivery drones on the other hand have attained high technology readiness level and are being operated to provide services across the globe (e.g. Manna, Wingcopter, Airhub, Zipline). In some Swedish counties and municipalities, drone manufacturers and drone service companies have also emerged to provide demonstrations, and delivery services (e.g., Katla, Everdrone, Airoloit). However, the market is volatile with companies emerging and dissolving quickly (e.g. Aerit).

Despite rapidly improving technological readiness in Europe and Sweden, knowledge on civilian drones remains largely limited on its impacts, governance, and societal uncertainties, particularly at the local and regional scale. Additionally, it is least explored within the broader context of sustainable transport transition, which is essential to guide decision-making that determines future transition pathways.

Sustainable transport is a cross-sectoral challenge. Globally, the transport sector is the second largest energy consumer, accounting for approximately 29% of energy consumption in 2023 (IEA, 2025), and in Sweden the transport sector is the third-most energy using sector of about 78TWh in 2023 (Energimyndigheten, 2025). The surge in demand for e-commerce and fast delivery times exacerbates this trend, particularly in the last-mile delivery segment, where trucks are often partially loaded, contributing to poor environmental impacts and higher costs (Jaller et al., 2021). In Sweden, the government aims to achieve zero net emissions by 2045 through a fossil-free transportation system and with a 70% reduction (compared to 2010) in domestic transport emissions by 2030 (Trafikverket, 2025). Such ambitious targets are also tied together with improving accessibility, connectivity, and provision of equal access to sustainable transportation in rural or low population density areas, as emphasized by the European Green Deal (EU, 2025). Meeting such multifaceted objectives thus often involves experimenting with alternative mobility technologies that possess the potential for mobility planning (Hoogma et al., 2005).

Electric mobility is a major energy transition strategy in transport (Collazos et al., 2024). Emerging research often explores this as an opportunity for the use of drones, along with addressing transport challenges such as tools for improving accessibility, relieving congestion, and improving service provision in rural contexts (Garg et al., 2023), which are significant factors for sustainable transport (Banister, 2008). However, emergence of a new technology within existing systems and its ongoing transitions have complex transitions dynamics that need to be understood to support multi-dimensional decision-making (Geels et al., 2017a), not limited to barriers and drivers for their implementation as critiqued in Rip and Kemp (1998), but also for its own merit related to social, environmental, economic consequences, and their systemic interactions. The impacts, risks, and uncertainties that technologies present however should be evaluated and addressed through governance practices such that the transitions that unfold with their adoption can be effectively managed (Rotmans et al., 2001). Thus, this thesis places civilian drones within the broader context of transport transition, and sets out to add knowledge for decision-makers (primarily within policymaking, transport and drone industry) and researchers about the possibilities and challenges drones would bring forth for

low-carbon transition and provide indications regarding what is necessary to manage the emergence of this technology in line with achieving sustainability goals.

1.1 Aim and research questions

This doctoral thesis aims to explore the emergence of drone transport within the broader transition towards a sustainable transport system for low-carbon and energy-efficient transport futures. By studying the interplay of policy aspects, planning considerations, public acceptance, energy and emission impacts of drones, the thesis seeks to understand the potential of drones towards sustainable transport transitions, identify barriers and drivers that would influence decision-making, and governance approaches for the development and implementation of drones purposefully towards meeting transport needs locally. Accordingly, the following research questions are addressed:

RQ 1: What impacts and potential roles can civilian application of drones have in contributing to sustainability transitions within the transport sector?

RQ 2: What barriers and drivers influence the integration of drones for energy efficient transport transition?

RQ 3: How to address the emergence of drones to meet local sustainability needs?

1.2 Scope and delimitations

This thesis considers drone as an emerging technological innovation, with fo-cus primarily on transportation. Drone use in the context of military applica-tions, or commercial applications besides transport, like surveillance and mon-itoring, photography, agriculture and forestry, etc (Aydin, 2019) are excluded. How drones fit in the efforts to achieve sustainable, low-carbon, energy-effi-cient transportation is of interest in the thesis. Therefore, the thesis does not examine detailed technical or engineering aspects of the drone or its corre-sponding infrastructure like vertiports or airspace; instead, technological and infrastructural aspects are discussed in relation to sustainability policy and planning considerations.

Geographically, the empirical focus is delimited to European, and in some studies specific Swedish context. Developments in European Union policies, together with Swedish national, regional, and local policy landscapes, and as-sociated actors form the main institutional context. The results and insights from the thesis are, however, considered to be applicable in the other geograph-ical contexts with comparable scenarios and institutional arrangements.

All appended papers in this thesis are synthesized and analysed to address the three research questions, drawing on both quantitative and qualitative re-search. Quantitative assessments consider drones within the last-mile delivery due to its higher feasibility and relevance, while energy assessment is limited

to include energy use in the use-phase of last-mile delivery. Broader environmental consequences, like noise, effects on biodiversity, land-use, etc are discussed qualitatively. Infrastructure is excluded from quantitative analysis, while it is addressed qualitatively. Public perspectives are taken from previous research in the field. Finally, the thesis acknowledges the significance of behavioural aspects needed for sustainable transport transitions (Richardson, 1999), however, these aspects are not the focus of analysis.

2. Emergence of drones for transport transition

Use of drones for civilian purposes in many parts of the world is mostly within experimental phase. In Sweden, some projects have explored the integration of drone technology into sustainable transport systems. One such project focused on using drones to deliver defibrillators in emergencies. Led by the Västra Götaland region, this project aimed to reduce response times in cardiac arrest cases by deploying drones equipped with defibrillators (forskning.se, 2023). The study highlighted that the chances of survival increase significantly if a defibrillator arrives ahead of emergency medical personnel (VGRFokus, 2021). Similarly, projects have explored the use of drones for transporting blood bags between hospitals, showing reduction in travel time compared to ground transport (Greenflyway, 2022). Within the delivery segment, particularly last mile delivery has received significant attention. Faster spare part delivery using drones have been explored where comparisons have been made with ground transport with concerning delivery time (SVT, 2022), and delivery of goods. Another project has explored drone-based food transport in the Stockholm archipelago with grocery deliveries via drone demonstrating potential applications for rural food logistics (Sälstrand, 2022). Likewise, Torsby municipality conducted a pre-study which shows the demand for home deliveries from residents in rural areas for improved access to essential goods in sparsely populated areas (Kommun, 2022), followed by a pilot project for home deliveries which showed challenges related to regulations, cost-effectiveness, digital connection, etc (Kommun, 2024). Another project, with the involvement of Swedish actors, explored the integration of urban air mobility into emergency medical services by testing drones for transporting medical personnel and supplies in European cities (AIRMOUR, 2024). These examples collectively highlight the emergence of drones and interest in its exploration within Sweden's sustainable transport transition, primarily to improve transport efficiency and accessibility across various sectors, while the low-carbon potential, although prevalent in discourse (SAAB, 2024; Vattenfall, 2021), is less explored.

The following sub-sections will explore drone technology, its operational infrastructural needs, and previous research regarding environmental and social aspects. It is important to note that the intention of the following sub-sections in not to conduct an exhaustive review, rather, to clarify what is meant by drones in this thesis, what technical and regulatory complexities infrastructure planning for drones can entail, and what current state of research indicates regarding its impacts. Furthermore, these subsections also highlight the research gaps identified that are subsequently addressed in this thesis.

2.1 Drone technology

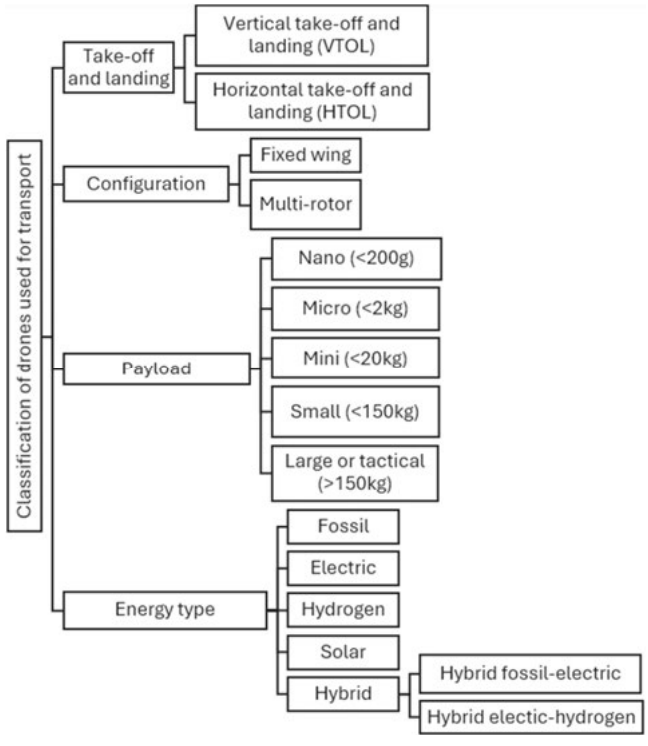


Figure 1. Classification of civilian drones used for transportation

Drone technology for civilian transport applications varies in size and configuration. There is not a specific way of classifying and distinguishing these drones since this technology has not yet reached a dominant design in the market. Hence it is evolving and new concepts from different manufacturers emerge. What does remain consistent with that of military applications are how these drones can be classified based on their landing functionality, configuration, payloads, and energy source (as shown in Figure 1). Based on how they land, broadly two types of drones can be found- (a) Vertical Take-Off and Landing (VTOL) which, like helicopters, take-off and land vertically; and (b) Horizontal Take-Off and Landing (HTOL) which have fixed wings, and like conventional aircrafts they require a runway for take-off and landing and are more energy efficient (Darvishpoor et al., 2020). VTOLs commonly are designed with propellers for vertical thrust, and are called X-copters, which are considered to be the most responsive constructions (Sivakumar & Tyj, 2021). "X" denotes the number of propellers, such as duo-, quad- hexa- or octa-. Drones also come with varied payloads, typically a few hundred grams, to very large payloads, which can be a few hundred kilos. Based on payloads, they can also be classified as nano, micro, mini, small, medium, large (Sönmez et al.,

2022). Based on energy source, drones can be classified as fossil, electric, hydrogen, solar-powered, and hybrid concepts (Townsend et al., 2020). However, electric vertical take-off and landing vehicles (eVTOLs) are the most referred to drones in discourses.

2.2 Infrastructural requirements for drone operations

2.2.1 Airspace infrastructure

Drones are suggested to operate in the lower airspace above 400ft (120m), generally between 1500 ft (450m) and 4000 ft (1200 m), depending on the drone itself, the type of application, and where the operations are conducted it could vary 300 ft from ground or 150ft from the tallest building with 6500ft of maximum altitude (Bauranov & Rakas, 2021; Qu et al., 2023). Drone transport operations will require access to very low-level airspace (VLL), to take-off, land, deliver, etc. The permissible altitude range for drones considers factors such as safety, airspace congestion, and airspace utilization efficiency. There are many different concepts of how airspace designs to create 3-dimensional corridors for flight planning in the form of layers, cells, zones, tubes, and lanes (Bauranov & Rakas, 2021; Sunil et al., 2015; Sunil et al., 2016).

In Sweden, airspace below 2900 m is considered the lower airspace (LFV, 2023). Within this lower airspace, new services are being developed under the initiative of European Union Aviation Safety Agency (EASA) called the "U-space" framework to support the growing demand for drone operations while ensuring safety, efficiency, and interoperability (Transportstyrelsen, 2025). U-space services are designed to be highly automated and scalable, allowing for seamless integration with existing air traffic management (ATM) systems (EASA, 2021a). U-space services progress through four levels where U1 services, provided since 2021, establish the foundational infrastructure with e-registration, e-identification, and geofencing, ensuring regulatory compliance and basic monitoring. U2 services, provided since 2023, introduce operational functionalities like flight planning, approval, tracking, and integration with air traffic control for coordinated and authorized drone movements. U3 services, provision from 2027, are provided for advanced airspace management with conflict resolution and detect-and-avoid systems, enabling safer operations for high drone traffic. At the highest level, U4 services are said to enable full automation and integration with air-traffic management and manned aviation. These services are introduced in a phased approach, progressing from foundational capabilities to advanced automation and high-density airspace management (Barrado et al., 2020).

Since 2023, U-space service providers can get certified by the national transport authority to provide these services privately and offer digital services to drone operators, facilitating safe and efficient flights. A Common Information Service (CIS) acts as a central hub for sharing critical airspace data, ensuring that all U-space participants have access to up-to-date information (EASA, 2020). Traditional air navigation service providers interface with U-space to maintain coordination between drone operations and existing ATM

systems. Additionally, regulatory authorities oversee U-space operations, enforcing compliance and ensuring that safety standards are met.

In Sweden, the Swedish Transport Agency is the responsible authority with Swedish air navigation service provider (Luftfartsverket) seen to conduct analysis for airspace planning. Besides the implementation of frameworks and regulations from international regulatory bodies such as the International Civil Aviation Organization (ICAO), EU, and national transport and aviation authorities, the standardization of the airspace will require involvement of regional and municipal actors (also addressed as local actors in this thesis) will have to engage in the topic through planning processes, and aligning them with local needs (EU, 2022b). However, how to do so is not addressed in current state of research, a gap that has been taken on in this thesis.

2.2.2 Ground infrastructure

Drone operations require ground services such as space for take-off and landing, communication, charging, ground handling, maintenance, etc. Infrastructure on ground has to be integrated with both the airspace and existing built infrastructure. Take-off and landing of drones entail selection of sites which provides appropriate conditions for such operations with low risk. Drone fleets will require central landing sites called vertiports, which can be seen as airports for drones. As addressed in EASA's document outlining "*prototype technical specifications for vertiport design*", technical standards and regulations for vertiport prototypes are in the early stage (EASA, 2022). Due to the absence of specific guidelines for vertiport layouts, most research relies on existing heliport regulations, given their structural similarities, particularly reference standards set by the Federal Aviation Administration (FAA) in "AC 150/5390-2C: Heliport Design" (2012) and the International Civil Aviation Organization (ICAO) in "Annex 14 – Aerodromes – Volume II – Heliports" (2013) (Brunelli et al., 2023).

Based on this, a typical vertiport pad structure generally consists of three zones, a safety area (SA) surrounding a final approach and take-off area (FATO) which in its turn surrounds a touch-down and lift-off area (TLOF) as shown in Figure 2.2 (Brunelli et al., 2023; Taylor et al., 2020). The TLOF is a paved area used for the drone's take-off and landing where the minimum length and width should correspond to the maximum dimensions of the vehicle (Brunelli et al., 2023). FATO is the area where the final phases before landing and initial phases after take-off is performed, for example hovering (Taylor et al., 2020). The size of the FATO should be at least twice as big as the maximum dimensions of the drone (Taylor et al., 2020). The SA has the purpose of reducing the risk of accidents and has half of the maximum dimensions of vehicle bigger than the FATO (Taylor et al., 2020). According to EASA specifications, the vertiport operator provides a layout plan with the dimension of "*the elevation of the vertiport, FATO, TLOF, SA, clearway, distance from the vertiport edges to the perimeter of SA or clearway, and approach/departure paths showing locations of buildings, trees, fences, power lines, obstructions (including elevations), schools, places of worship, hospitals, residential areas, and other*

significant features" (EASA, 2022). Additionally, vertiports have gates, where the drones will be parked, and taxiways, as shown in Figure 2.

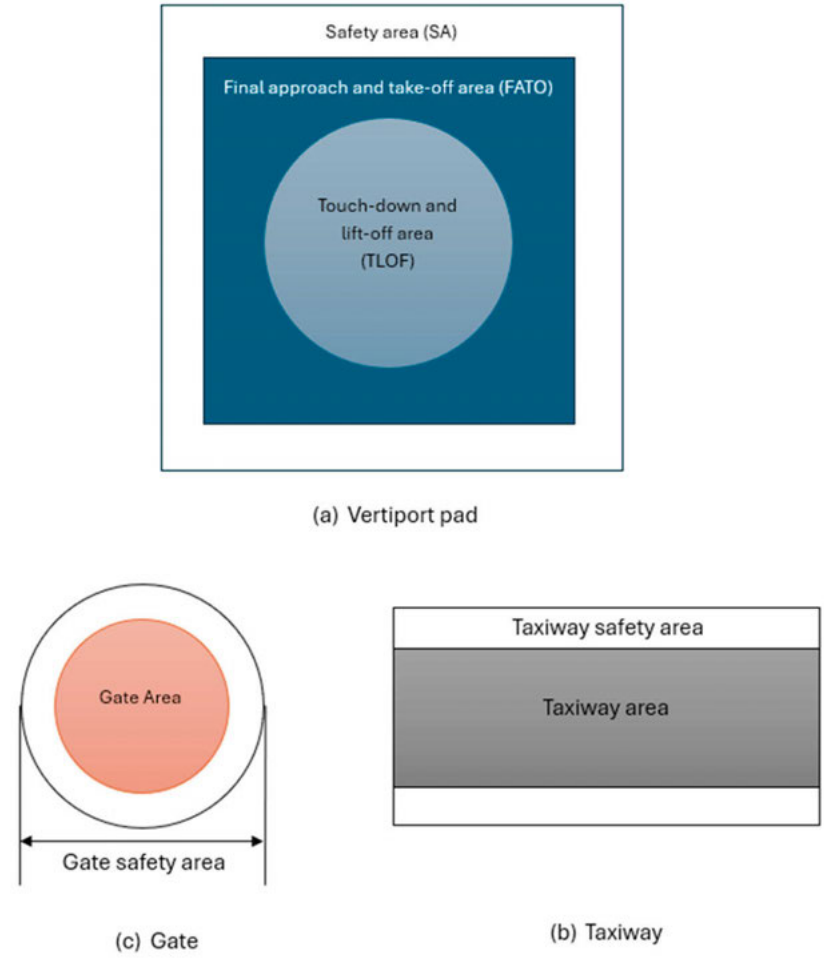


Figure 2. Standard design of a vertiport (Taylor et al., 2020)

Using this as a starting point, studies have developed many vertiport layouts, similar to helipad designs, where there is not one ideal design as it varies depending on space availability (Brunelli et al., 2023). However, irrespective of the vertiport type, the airspace surrounding the vertiport is safeguarded for safe flight take-offs from the vertiport. Obstacle limitation surfaces (OLSs) and obstacle-free volume (OFV) (see Figure 2) describe the airspace around vertiports to enable safe and obstacle-free drone operations (EASA, 2022). Specifically, OLSs define the airspace around a vertiport that must remain free of obstacles to ensure safe aircraft operations during approach, landing, take-off, and departure (EASA, 2022). The OFV is a funnel-shaped area above the

vertiport designed to remain entirely free of obstacles. It provides additional safety for drone operations by accommodating their ability to perform vertical take-offs and landings with significant vertical segments (EASA, 2022).

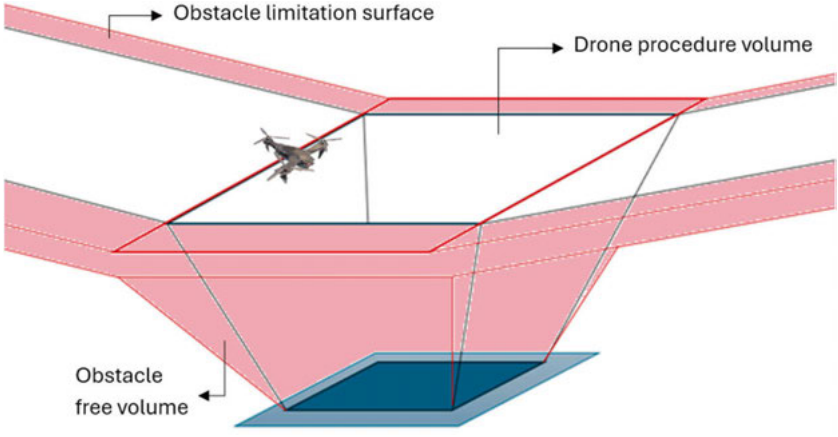


Figure 3. Airspace design surrounding a stationed vertiport (EASA, 2022)

Determining optimal positioning of the vertiports is addressed is said to be more critical than the number of vertiports (Syed et al., 2017) This is considered more complex than traditional transportation planning due to varying performance of different drones, uncertain air traffic, and fluctuating travel demand (Yan et al., 2024). Integrated optimization models are generally developed for planning vertiport locations and networks accounting for site selection, operations, and demand while considering factors like passenger behaviour, travel patterns, battery logistics, and queue management to enhance overall transport efficiency and aiming to maximize users, revenue, and vertiport capacity (Kai et al., 2022). Key considerations also include accessibility, connectivity with roads and public transit, and land use compatibility (Volakakis & Mahmassani, 2024). While much of the research has focused on concepts and simulations, studies highlight the need for strategic vertiport placement that balances travel efficiency with environmental impact, and societal costs (Mavraj et al., 2022; Rahman et al., 2023; Rothfeld et al., 2021).

Studies also suggest repurposing existing infrastructure such as to be multipurpose vertiport locations such as parking structures, rooftops, and underutilized transport hubs (Volakakis & Mahmassani, 2024) and airports to address zoning challenges and enabling effective space utilization. Off-shore floating vertiports are also designed to address spatial constraints and integration with maritime (Kim et al., 2024). Addressing such complexities creates high requirements for space efficient designs (Rimjha & Trani, 2021).

Furthermore, digital infrastructure is another critical part of ground infrastructure for the implementation of U-space services, traffic flow management, conflict resolution, separation in flight, trajectory planning, flight operations

and ticket distribution, ground transport coordination, autonomous ground vehicles, communication and navigation, data security, simulation tools, etc (Liu et al., 2024; Yan et al., 2024). In addition to this, energy infrastructure plays another critical role in establishing drone operations. Studies highlight the critical impact of battery fast-charging and swapping on effective drone transport implementation (Ratei et al., 2024). Taye et al. (2024) shows that the energy demand at a vertiport is primarily influenced by the number and timing of departing flight missions (Taye et al., 2024).

2.3 Previous research on social- environmental uncertainties

2.3.1 Energy and environmental challenges

The environmental impact of drone transport is indeed a concern, particularly as these vehicles rely on electricity, which may still contribute to emissions depending on the energy sources used (Afonso et al., 2021). While electric propulsion reduces direct emissions compared to fossil fuel-based transportation, energy consumption during drone operations is said to remain significant. Drone transport is speculated to demand substantial energy resources, potentially increasing the overall transportation volume due to enhanced convenience, leading to greater energy consumption (Ahmed et al., 2020).

Since drone transport is highly dependent on the electricity grid, the environmental benefits of using these vehicles largely on the energy mix of the grid (Filippone & Barakos, 2020). The type of energy utilized not only affects emissions but also influences vehicle performance, operational efficiency, noise levels, design considerations, and public acceptance (Lewis et al., 2021). The variability in energy sources further complicates sustainability assessments, making it imperative to evaluate drone transport from a life-cycle perspective. For instance, while batteries are a crucial component of these vehicles, their production, maintenance, and disposal require significant energy and critical materials (Filippone & Barakos, 2020). During drone use, although their energy demand during cruise is optimized, they have high energy demand during take-off (Figliozzi, 2017; Kasliwal et al., 2019) and hover (Kirschstein, 2020). Additionally, production of drones require materials such as carbon fibre composites (for the airframe), permanent magnet (for the motors), aluminium, electronics, lithium polymers (for batteries) which can contribute to high emissions (AIRMOUR, 2023).

Operational inefficiencies also pose environmental challenges. Drones used for transporting goods or passengers may frequently travel in one direction fully loaded and return empty, leading to unnecessary energy expenditure and emissions, requiring optimization of operations to minimize environmental impact (Rimjha et al., 2021). Furthermore, different operational models, such as private ownership, shared services, or hybrid systems, will influence the overall impacts of drone transport (Ahmed et al., 2020). Therefore, studies suggest comprehensive environmental regulations to account for emissions, energy sources (Lewis et al., 2021), broader transportation system integration (Ahmed

et al., 2020), and logistical considerations (Rimjha et al., 2021). In this regard, drone transport is suggested to contribute likely to emissions and energy efficiency regulations similar to those applied to conventional fossil-based transport modes.

Impact assessments or energy and emissions evaluations are still in an early phase where a multitude of assumptions regarding possible scenarios dictate results. Also, methods for life-cycle assessments for drones from cradle to grave including production and recycling is still developing, and assessments for the use-phase has been more common. Studies usually develop optimization problems (Dukkanci et al., 2020) for drones-only, or develop drone and truck problems through which their emissions are optimized (Baik et al., 2025; Bruni et al., 2023), and not evaluated. In evaluations, some studies are optimistic and show small quadcopters to have lower emissions per package than diesel or electric trucks and large drones to have lower emissions than diesel trucks (by 34%) and that of personal vehicles (Stolaroff et al., 2018). Rodrigues et al. (2022) present with experimental flight data that small drone could cut emissions per package by about 94% for small parcels in low-carbon electricity systems (Rodrigues et al., 2022). However, the same study also shows very high energy usage per ton-km basis (supplementary, S1 in Rodrigues et al. (2022)).

Other studies show a more differentiated picture where drones could increase emissions depending on transport options and routes (Figliozzi, 2017; Kirschstein, 2020). Some studies conclude that environmental benefits of drone deliveries are greater in areas with low population densities (Park et al., 2018). Yowtak et al. (2020) presents life-cycle assessment of drone- truck combined models which shows that that drone-truck combinations are better than diesel vehicles, whilst being worse than electric vehicles in most cases (Yowtak et al., 2020). However, Baldisseri et al. (2022) conducts life-cycle assessment of drone- truck combined model in the very last mile where trucks play the role of mobile depots (synchronized model), which results in such models having lower emissions than diesel and electric trucks (Baldisseri et al., 2022). Drones are, however, suggested to have an advantage over trucks in service zones that are closer to the depot or have fewer recipients, while trucks are more efficient for distant, high-demand areas, thus recommending a hybrid system where drones handle nearby deliveries and trucks conduct farther deliveries to minimize emissions (asynchronized model) (Goodchild & Toy, 2018). These studies are seen to consider nano and micro drones (based on the classifications presented in Figure 1), and not small or large drones which is a key gap that was identified and addressed in this thesis.

Beyond direct vehicle emissions, infrastructure requirements also contribute to environmental concerns. Although relative to land-based modes of transport, land needs for drones are suggested to be low, the need for extensive support infrastructure, such as vertiports, warehouses, etc could lead to considerable resource consumption if not carefully managed (Kellermann et al., 2020; Straubinger, 2019). Moreover, drone transport has the potential to reshape urban development patterns. Increased accessibility may encourage individuals to move away from urban centres while still maintaining

connectivity, potentially exacerbating urban sprawl and leading to higher travel distances and inefficiencies within existing infrastructure (Rendón et al., 2021; Straubinger, 2019). There is also a lack of understanding of the consequences of this mode of mobility to diverse species, especially those that are dependent on the lower airspace and reduction of caution to other human threats (Kellermann et al., 2020).

2.3.2 Social challenges and public acceptance

Public acceptance is regarded as one of the major challenges for integrating drones in the lower airspace as a part of the ground-based transportation system as well as conventional manned aviation. Many surveys have been conducted in Europe, USA, and Asia previously to establish the acceptance rates and identify potential factors that affect community acceptance, which are generally related to noise, visual pollution, environmental impacts, privacy, safety, security, costs, etc.



Figure 4. Public concerns identified in literature (from Paper III)

In (Yedavalli & Mooberry, 2022), a survey was conducted with 1540 respondents from Los Angeles, Mexico City, New Zealand, and Switzerland. Results revealed that 44.5% of respondents supported or strongly supported UAM and that 41.4% of 114 respondents thought drone transport was safe to very safe. A meta-analysis from Legere of former US public surveys (Legere, 2019) and a survey conducted in Germany (Eißfeldt & Biella, 2022), addressed use of small drones in emergency services as police surveillance, search and rescue. The surveys showed acceptance levels of 60% and 49% respectively (Eißfeldt & Biella, 2022; Legere, 2019). The meta-analysis focused on the different acceptance levels per mission, with public missions having higher acceptance than private/commercial uses (Legere, 2019). The survey in Germany shows some major concerns to be misuse of drones for crime and violation of privacy (Eißfeldt & Biella, 2022).

A survey conducted in Singapore in 2020 with 1000 respondents showed an average acceptance of 62% (Tan et al., 2021). In 2021, EASA conducted a survey with 4000 residents of six European cities namely Barcelona, Budapest, Hamburg, Milan, Paris and Öresund along with qualitative interviews with

focus groups of local, national, and European stakeholders. (EASA, 2021b). This survey obtained the highest acceptance of 83%. Social acceptance factors for drones are suggested to be the same as that for other smart technologies (EASA, 2021b).

A survey conducted in South Korea with more than 1,000 citizens and 44 experts shows the willingness to use drones (both passenger and cargo) was 47%, and decreasing when the automation of the vehicles increased (Park, 2021).

In 2022, Nordic survey of public acceptance for drone transport was conducted at VTT Technical Research Centre of Finland, Tampere, with 1000 citizens, out of which 50% of participants were from Sweden and Finland respectively which showed higher acceptance particularly for improving accessibility in hard-to-reach areas (76%), emergency services (almost 100%) and home deliveries to the backyard or private space (40%) (Oksman & Kulju, 2022).

Smith et al. (2022) suggests that the level of awareness and knowledge of the technology influences the perceived risks by citizen and stakeholder groups. The *familiarity hypothesis* suggests that greater knowledge and experience with drones generally lead to lower perceived risks and higher acceptance (Komasová et al., 2020) However, Komasová et al. (2020) also found that effects of learning about drones is non-linear, as initial awareness may heighten risk perception, and gradually, deeper understanding eventually reduces concerns. However, this is also dependent on personal experience with the impact of the technology, as is the case of noise where after experiencing the sound of a drone, noise concerns can increase (Eißfeldt, 2020).

It is seen that advantages of drones such as delivery speed, which is said to save time for about 45% of the commuters in case of passenger mobility (Bulusu et al., 2021), and environmental friendliness are core drivers for drone adoption (Kellermann et al., 2020). Further, strong connections between social and economic benefits and public acceptance can be seen. Cost efficiency, translating to cheaper costs for users and strong service reliability is an important aspect (Al Haddad et al., 2020).

Research on risk mitigation indicates that trade-offs exist between addressing public concerns (Çetin et al., 2022). Public acceptance literature on drones has widely focused on understanding public willingness, concerns, and modelling public attitude (Kellermann et al., 2023) primarily to understand how they can affect integration of drones. Far less attention has been given to on how public concerns can be incorporated in decision-making, or how they interact with planning aspects, which has been another critical aspect incorporated in this thesis.

3. Theoretical perspectives

Technologies can be radical and disruptive, or incremental. Radical technologies are said to require varied technological capabilities in contrast to incremental technologies that are dependent on the existing and well-established ones (Christensen, 2015). The radicality of a technology may be seen in one dimension and its incrementality in another, such as introducing a new technological paradigm yet inducing incremental changes in its impact on the economy and vice versa (Lundvall, 1992). Nonetheless, technologies can be understood as components of broader "socio-technical systems" such as energy and transport. Such systems conceptualize the complex interplay of technologies with infrastructures, institutions, policies, markets, scientific knowledge, and cultural practices, which co-evolve with technological transitions (Geels, 2004; Smith et al., 2005). Emergence of automobiles brought forth development of new infrastructure like roads and energy supply, new services, new regulations, changed user practices, impacted everyday how people lived and worked, as well as policy and planning approaches, thus exemplifying broader systems changes with regard to both structure and society (Markard et al., 2012). Therefore, it is important to understand emerging technologies with a socio-technical view to understand their systemic impacts through their complex interactions and guide their development.

The Multi-Level Perspective (MLP) on socio-technical transitions is one such framework that helps in understanding how technologies emerge, evolve, and change existing systems as non-linear processes influenced by developments and complex interactions on multiple systemic levels (Geels, 2002). Therefore, in this thesis, MLP is used as a heuristic and analytical framework for studying the dynamics with emergence of drones within sustainable transport transition.

3.1 The multilevel perspective

The MLP framework conceptualizes the dynamics involved in socio-technical systems transition such as transport and energy into three analytical levels- namely, niches on the micro-level, socio-technical regimes on the meso-level and socio-technical landscapes on the macro-level (Geels, 2024).

Niches are said be emerging in "protected spaces" where innovations with varying degrees of maturity and radicality (or deviations from existing rules) emerge, often through research and demonstration projects due to external trends (Geels, 2024; Geels, 2004). This could vary from radical technical innovations, social innovations, business model innovations, and infrastructural innovations (Geels, 2024; Geels et al., 2016). Some novel technologies could persist to develop dominant designs and gain internal momentum through

establishing visions, creating social networks, and learning processes (Geels, 2002) that would determine their persistence into the regimes.

Socio-technical landscapes are the broader context influencing the niche and pre-existing regime dynamics, which make up our world through physical aspects such as infrastructure, to abstract ones such as macro-economic trends, political ideologies, societal beliefs, etc (Geels, 2012).

Socio-technical regimes exist within this landscape. The regimes comprise of socio-technical systems which are the physical structures/artefacts, actors (including social groups) who maintain and reproduce the links in the system, and rules that guide and orient the activities of actors (Geels, 2005). On a meso-level, they are patchwork of regimes or different socio-technical constellations, constructed by previous adoption and diffusion of technologies, preexisting networks of institutions, practices, and technologies that try to sustain existing development paths while shaping and reinforcing socially valued functions (Geels, 2002; Rip & Kemp, 1998; Smith et al., 2005). An ideal-typical process of systems transition is said to unfold when niches that have gained internal momentum take advantage of windows of opportunities that have been created in the regime through destabilization or "*tensions and mismatches*" that can occur due to exogenous pressures from the landscape (Geels, 2004). As new technologies emerge from niches within regimes, they co-evolve with these functions, reflecting entrenched beliefs and interests in practices, and hence it is characterised by incremental changes that enable stable systems determined by path dependency and lock-ins.

3.2 Path dependence and lock-ins

Path dependency refers to the process where past decisions and developments shape the trajectory of future choices in the regime, often creating a reaction when a change is introduced. As Pierson (2000) notes, path dependency invites recognition of "*the causal relevance of preceding stages in a temporal sequence*". This occurs due to several self-reinforcing mechanisms such as high exit costs- which makes changing course have high economic liability, learning effects- that increases efficiency with familiarity, coordination effects- where widespread adoption enhances value, adaptive effects- where self-fulfilling expectations that reinforce the dominant choice based on anticipated decisions (Pierson, 2000); which provides positive feedback loops and thus strengthens existing patterns and norms in the regime (Kemp et al., 2007).

Geels (2019) suggests that niche innovations face significant barriers within existing regimes due to multiple lock-in mechanisms, namely techno-economic lock-ins from sunk investments in infrastructure, economies of scale, and decades of performance improvements, making established technologies more cost-effective and resistant to change; social and cognitive lock-ins further reinforce stability through ingrained routines, dominant mindsets, and social capital built around long-standing technological practices, limiting openness to alternative solutions; political lock-ins play a crucial role, as existing regulations favour incumbent actors, while lobbying efforts by vested interests work to maintain the status quo and obstruct radical innovation (Geels, 2024;

Geels, 2019). In addition, Pierson (2000) highlight the role of discursive lock-ins, or unchallenged assumptions, which further entrench existing trajectories. Deterministic views of technology have been contested by historical institutionalists suggesting that longevity of adopted niches is dependent on institutions which require active reproduction and political sustenance post-choice (Thelen, 1999). Power and conflict play an instrumental role in regime actors holding on to "discursive, material and institutional forms" in resisting change (Geels, 2014). Buschmann and Oels (2019) introduce the concept of path creation, highlighting moments when actors actively facilitate institutional change. These moments could represent windows of opportunities for breaking away from existing trajectories and setting new directions. Thus, while historical choices strongly influence the choices made for the future, agency and critical decision points alter the course of development, allowing the emergence of radical change from innovative networks operating at the margins of dominant regimes (Smith, 2007).

3.3 Moving from emergent towards purposive transition

Systems transitions are said to unfold in many ways, which Berkhout et al. (2004) conceptualizes into four based on degree of actor coordination and resource availability:

1. endogenous renewal, referring to planned change driven by actors within the existing regime using their own (internal) resources,
2. reorientation of trajectories, that occurs when internal or external shocks trigger regime actors to adjust and respond from within,
3. emergent transformation develops through uncoordinated external pressures, often led by small or new entrants,
4. purposive transitions, which involve intentional, coordinated efforts originating outside the regime to steer change in line with explicit societal goals (Berkhout et al., 2004).

"Sustainability transitions are defined as long-term, multi-dimensional, and fundamental transformation processes through which established socio-technical systems shift to more sustainable modes of production and consumption." Pg 956, (Markard et al., 2012). Low-carbon transition strategies are said to require solutions outside of existing regimes (Geels, 2024) and deliberately driven towards established goals and visions, thus necessitating purposive transitions (Geels et al., 2017a).

This perspective is crucial in guiding decision-making processes for an emerging technology towards meeting societal goals that are predicated on sustainability transitions. Such complex transitions towards an unknown future deliberates resilience thinking (Folke et al., 2010). At any given moment, selection pressures exist on regimes, such as niches developing from an existing regimes or regimes across sectors, creating competition, alternative societal visions, landscape trends, or public discourse challenging existing regimes (Smith et al., 2005). The capacity and resources of the regime to deal with these

selection pressures is called adaptive capacity, where the actors respond to potential disturbances, to take advantage of opportunities, or to respond to consequences (Folke et al., 2010; Smith et al., 2005; Walker et al., 2004). Transformative capacity of the regime in contrast is referred to as "the capacity to create a fundamentally new system when ecological, economic, or social structures make the existing system untenable" (Folke et al., 2010; Walker et al., 2004). It can be inferred that adaptive capacity is the ability actors and institutions possess to accommodate, adapt, and overcome disturbances to return a regime to its normative state, while transformative capacity is the ability of the socio-technical system to take new transitional paths beyond the norms, which is essential for purposive low-carbon transition. Thus, this thesis takes on this perspective to explore drones within the sustainable transport transition context.

3.4 Policy and planning

Openness to fostering innovation, through continuous processes of trial and error, and learning are influential in their evolution and use windows of opportunity to address broader sustainability goals (Geels et al., 2017b). However, it is necessary to tread with caution and critique as emerging technologies have significant uncertainties regarding their actual emission reduction potential, patterns of resource use, and systems impacts such as rebound effects, environmental risks, etc., many of which are difficult to assess in the early stages of development. A growing body of research points to the limitations and risks of techno-optimism, which fails to consider systemic interdependencies, increase social-environmental inequalities, and produce unintended consequences (Jasanoff, 2016) that has led to problems like climate change where urgent policy interventions are imperative (Levin et al., 2012). Here, it is critical to understand that what is considered sustainable can change over time (Markard et al., 2012).

Generally, policy arrangements can be understood through vertical and horizontal dimensions, where vertical entails complex interactions between "different policy sectors and/or levels of administrative government", and horizontal which entails relatively simpler interactions between different instruments, policies, and goals within the same policy sector and level of government (Howlett & Del Rio, 2015). Transition management aims to address science, innovation, and sectoral policies by integrating vision development, agenda building, practical experimentation, and reflexive learning through multi-level (vertical) policy framework (Kemp et al., 2007). Operationally, it involves four types of activities at different levels: strategic level where the transition arena enables problem structuring and the development of guiding visions; tactical level where coalitions and networks shape agendas and outline transition paths; operational level where experiments and projects test and demonstrate innovative practices; and continuous monitoring and evaluation ensure that both processes and outcomes are assessed, enabling strategies to be adapted and feeding back into broader learning processes over time (Kemp et al., 2007). This requires complex multi-policy mixes that are multilevel and multi-goal requiring signifying cross-sectoral vertical policies (Howlett & Del Rio, 2015).

Influence of incumbents in the regime and core alliances with policymakers can drive or hinder change based on vested interests in preserving the status quo (Geels, 2004). Discursive practices that mobilize collective sense-making and social construction of the transition, roles of existing or new artefacts, innovations, and behavioural changes addressing consumption is crucial yet often contradictory within climate policy for (Roberts & Geels, 2019; Wells et al., 2020). This informs visions of the future and tensions that also influences the success of a technology. Sense-making for emerging technology to extend to epistemic knowledge of the technology, its role within the systems, and what goals it supports (i.e., what does sustainable transition mean). Policy actors within the MLP often work with different values, ontologies, and vested interests and hence lead to conflicting understandings, competing interests, and power struggles (Geels, 2010). Drawing on Bourdieu's notions of doxa and habitus, policy actors are embedded in historically institutionalized practices and implicit framings, which influence how problems are framed, and solutions are legitimized (Bourdieu, 1977). Reflexivity is thus needed to be a part of the governance process (not only reactive or detached) (Loorbach, 2010) which can enable policy actors to critically assess the assumptions, values, norms, and institutional arrangements that shape practice.

Currently, the emergence of many alternative technologies and solutions has overwhelmed regimes, which need to foster urgency, commitment to radical transitions, support transition practices, and combine scientific evidence with creativity, cultural strategies, and systemic approaches (Loorbach, 2022). Leadership is therefore crucial to mobilizing policy priorities that are not just reactive. Geller (2003) suggests that while managers typically rely on extrinsic mechanisms through accountability, rewards, sanctions, and monitoring consequences, leaders actively foster responsibility, ownership, learning, and intrinsic motivation (Geller, 2003). Based on this perspective, proactiveness can be seen as essential for policy leadership.

Finally, building empirical evidence for sustainability and feasibility, and bridging science-policy gaps through inter- and transdisciplinary approaches, techno-economic and broader societal feasibility checks, while fostering the organizational capacities needed for guiding real-world transitions (Geels et al., 2017b; Loorbach, 2022).

4. Methodology

4.1 Research process

This thesis was the outcome of an industrial PhD which was a part of Future-Proof Cities research school funded by Knowledge Foundation (KK-Stiftelsen). Future-Proof Cities project set out to explore solutions through a co-creative approach with public and private companies to address relevant societal sustainability problems. In this context, my industrial PhD was conducted at Independent Business Group AB, which is a consultancy company for aviation infrastructure and had begun focussing on drone transport. Furthermore, my PhD studies was also affiliated with University of Gävle, with the Department of Building Engineering, Energy Systems and Sustainability Science, within the subject area of Energy Systems. It involved supervisors from different academic disciplines (energy systems, sustainability science, and industrial management). Thus, the research process of this PhD was inspired by a co-creative approach to develop transdisciplinary knowledge that is informed by scientific and societal practitioners.

Addressing complex real-world problems is said to require the integration of knowledge across disciplines, and the integration of tacit and explicit knowledge enabling research relevance towards addressing practical challenges from varied constructions of reality and epistemologies (Spangenberg, 2011). Transdisciplinarity is one such approach to research which allows knowledge co-creation across academic disciplines and by involvement of multiple actors to solve complex problems for sustainable transitions (Scholz, 2017). Science in this regard is considered a knowledge commons with a role to contribute to core societal development (Scholz, 2017); knowledge production of which features the involvement of stakeholders of the research issue (Bracken et al., 2015). It is defined as:

> *" a reflexive, integrative, method driven scientific principle aiming at the solution or transition of societal problems and concurrently of related scientific problems by differentiating and integrating knowledge from various scientific and societal bodies of knowledge." pg. 26-27, (Lang et al., 2012)*

A transdisciplinary co-creative process cannot be linear, rather it is cyclical and dynamic (Ranhagen, 2020) or iterative and recursive (Lang et al., 2012) and is highly contextual. Yet the three-phase ideal-typical method by Lang et al. (2012) provided a streamlined overview of the co-creation process (see Figure 5) that helped provide a framework for my research.

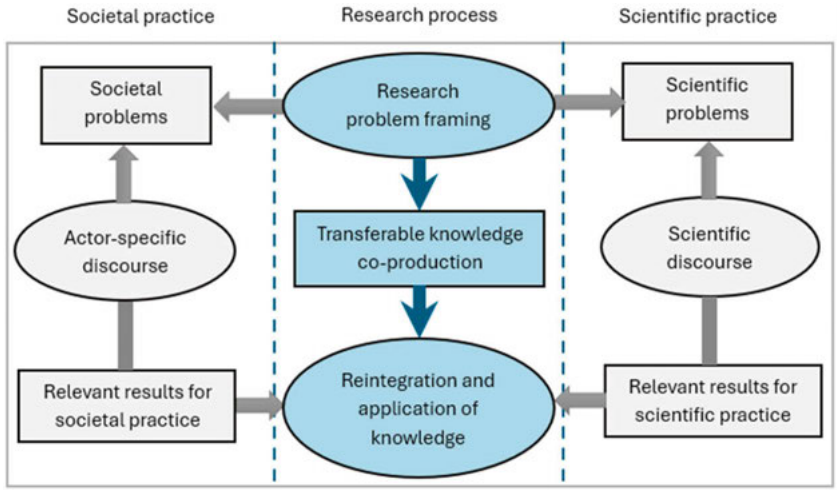


Figure 5. Transdisciplinary research process, inspired from Lang et al., 2012.

My PhD studies started with brainstorming sessions that were held with the supervisors from academia, mentors at the company, as well as joint sessions with both. These sessions occurred iteratively. Discussions with supervisors entailed scientific contexts of the emergence of drone technology. Co-creative discussions at the company and with the mentors unveiled societally relevant challenges for drone implementation with regard to policy and planning. These joint discussions shed light on the broader societal context and relevance of the research topic.

The policy aspect was thus taken on as the first part of the research, which led to Papers I and II. Simultaneously, a common understanding of an integrative process was created for knowledge creation, which evolved throughout my research journey. This included getting involved in company projects, enabling me to understand the practical challenges, acquire knowledge in the field, and integrate knowledge. Discussions with mentors helped identify the relevant niche and regime actors in the field. This included drone companies, aviation authorities, transport authorities, municipalities, regions, airports, logistics companies, and researchers. From this, relevant actor perspectives (particularly of policy and practice) were decided to be integrated through interviews. Further, a quantitative study was pursued for the impact assessment of drone delivery in Sweden for Paper IV. This was developed based on a demonstration project, "*Concept of Operations for European U-space Services - Extension for Urban Air Mobility*" (SESARJU, 2024), that I was a part of through the company. Already framed policy and planning challenges and findings from Papers I and II (such as the need for including public needs, involvement of regional planners, and transport actors), as well as potential trade-offs observed both through literature review, interview processes, and tacit knowledge developed, added to developing Paper III. Finally, outcomes from Paper IV regarding use of small drones in complementary ways with road-based

delivery vehicles, as well as input from logistics companies regarding challenges of truck loading, optimization, and infeasibility of truck-based delivery for low-density areas, provided the rationale to pursue another quantitative assessment in Paper V. Mentors at the company provided inputs from time to time (e.g., reviewing models, providing knowledge from practice) and supported knowledge integration in practice (e.g., developing training seminars and policy discussions).

4.2 Exploratory research design

This thesis dealt with an immature field of an emerging contemporary phenomenon, where prior research was quite limited. In such a situation, taking an explorative stance enables the researcher to explore the topic of interest and develop theories (Bryman, 2016), in contrast to testing a theory vis-à-vis the empiricist philosophy of logical deduction and hypothesis testing (Johansson, 2021). As Stebbins (2001) suggests, explorative research as an approach that allows one to be inquisitive, open-ended, inductive (where generalizations are drawn from data), become familiar with a subject, and investigate systematically (Stebbins, 2001). Such an approach is considered necessary to "*uncover new ideas and observations*" (Stebbins, 2001). Thus, initially, a broad scope for PhD research was developed, which involved a focus on drones as a mode of transport contributing to sustainable transport. Through the previously mentioned co-creative process, initial research needs in the field were explored, and a specific focus was built upon various findings throughout the research journey.

4.3 Overview of methods used

This thesis has used quantitative and qualitative methods to arrive at findings of the appended papers. An overview of the methods used in the appended papers and the research questions of this thesis that they contribute to is shown in Table 1 and described further.

Table 1. Overview of research methods used to answer the corresponding thesis research questions in appended papers

	Paper I	*Paper II*	*Paper III*	*Paper IV*	*Paper V*
Contributing to research question	RQ 2, RQ3	RQ 1-3	RQ 1-3	RQ 1-3	RQ 1-3
Type of research	Qualitative	Qualitative	Qualitative	Quantitative	Quantitative
Data collection	Interviews	Systematic literature review, Interviews	Integrative literature review, Interviews	Literature review, Interviews, Vehicle manufacturer data	Literature Review, Vehicle manufacturer data
Data analysis	Narrative analysis	Thematic analysis	Thematic analysis	Energy and emissions assessment Life-cycle costs Delivery time	Energy assessment

4.3.1 Literature reviews

Literature review is considered as an effective tool to critically analyse, summarise, and identify research gaps within existing knowledge regarding a phenomenon (Kitchenham, 2004). This formed the starting point of data collection in this thesis as the research area was dominated by technical knowledge of drone technology, drone flights, or infrastructure for drones, and lacked in sufficient scientific research related to policy, and risks. Literature review also helped as a starting point to gain an in-depth overview over the subject, identify research gaps (Paper II and III), develop a framework for the phenomenon (Paper II), and analyse various attributes of the drone system (Paper II and III).

A systematic literature review (SLR) was conducted in Paper II, using methods and guidelines inspired by Denyer and Tranfield (2009) and Durach et al. (2017). To ensure reliable results and transparency of the process, a five-step methodology was used where the research questions were defined, a search strategy was developed by selecting the reliable databases and key

search terms, inclusion and exclusion criteria of studies was employed, retrieving and manual selection of the literature, thematic analysis and synthesising the identified data, and finally reporting the results. For each step, thought was given to the aim and research questions of the study, deducing a pragmatic framework for this phenomenon, critical self-reflections, and the implicit immaturity of the field.

Potential trade-offs within planning for drone transport was identified during the SLR. However, it was outside the scope of Paper II. Thus, a subsequent study was conducted (Paper III) using the conducted SLR as a basis and further conducting an integrative literature review (ILR). This type of review helps collect literature from various aspects of an emerging topic to review, critique, and synthesize (Torraco, 2016). It is used to address concerns, identify research gaps, and offer preliminary conceptualization of the emerging topic (Torraco, 2016).

While the SLR was exclusively conducted for scientific literature, the immaturity of the field, small number of scientific studies, and the insufficient practice and policy context of drone transport implementation was the main rationale for including grey literature in ILR literature review (Benzies et al., 2006). The inclusion of industry knowledge on practical aspects of technological and infrastructural planning and operations was another important reason. Reliable grey literature were selected using quality assessment checks such as considering the credibility of the source, clarity of purpose, and outlet type that are suggested by Garousi et al. (2019). The grey literature also went through a thematic analysis process as the SLR. The SLR and the grey literature formed the basis of the ILR. A conceptual/thematic structure was developed from these outcomes, and more literature was collected for ILR with searches on the specific topics, and related literature with manual snowball sampling.

4.3.2 Interviews

Interviews were used in this thesis to effectively be able to gather inputs from relevant actors for the purpose of co-creating knowledge. This was also deemed critical due to immaturity of the research field (as noted from the literature reviews) in contrast to rapid developments occurring in technological developments and policies. Interviewing is a method of knowledge construction through interpersonal interactions, aiming to interpret the meaning of central themes and phenomena obtaining open nuanced descriptions (Kvale & Brinkmann, 2009). Interviews were used in this thesis as a primary mode of data collection for Paper I and for validation in Papers II and III. This thesis uses interview method by Kvale and Brinkmann (2009) to conduct the interview studies in a systematic and replicable manner.

As this thesis is explorative in nature, semi-structured interviews were chosen allowing the interviewer to use follow-up questions based on the interviewees' responses (Kvale & Brinkmann, 2009) and avoid limiting the responses of the interviewees (Bryman, 2016). The interview guide was then developed for and consisted of two parts:- a brief introduction and interview questions. The brief introduction consisted of stating who the interviewers were, corresponding affiliations, and the purpose of the interview. The interview questions were

open-ended and semi-structured. The questionnaire was discussed with other researchers in the field and modified based on their insights.

Interviewees were asked to recommend relevant persons who could potentially be suitable for the interview, actors were selected using snowball sampling (Bryman, 2016). Dominant regime actors from drone industry, aviation and transportation sector, national, regional, and local authorities were enlisted. From this initial list, interview guides were sent out to potential interviewees via email. Sending out the introduction and the questions beforehand was helpful for the interviewees to get familiarized beforehand with the interview purpose and be prepared for the questions.

A semi-structured interview guide was used, with the interviewer's judgment guiding the depth of exploration. The 'what' and 'why' questions were then followed up with 'how' questions. Each question from the guide was taken one by one, giving the interviewee sufficient time to unfold their stories and sometimes each of the planned questions was followed by some follow-up question depending on the interviewees' responses to emphasize more the main characters in the story or the main episodes. Microsoft Teams was used as a mode of communication to conduct the interviews.

First 18, interviews were recorded and transcribed (that were used for Paper I and II) to obtain concrete written material. The audio recordings were then transcribed using NVivo software. Later, the verbatim transcriptions were manually edited by the interviewers by listening to the recordings in comparison and keeping in mind the social interaction during the interview as recommended by Kvale and Brinkmann (2009). The interview scripts were then reviewed by the researchers and edits were made to fill in some missing details, and specific errors such as grammar and minor redundancies were removed from the transcript. As research progressed towards Paper III, two more interviews were conducted with an urban planner and drone service provider for validation. Notes were taken during these interviews.

Different methods are used for analysing interviews, depending on the focus on meaning (Paper II and III) or language (Paper I), thematic analysis and narrative analysis were used respectively. The interviewer's role as a co-producer and consideration of the original discourse and social context were given importance during analysis (Kvale & Brinkmann, 2009). While using interviews to validate the findings in Paper II and III, thematic analysis was conducted. In Paper I, where the interviews were the primary data, narrative analysis was conducted, which is elaborated further.

4.3.3 Narrative analysis

Transition management within complex adaptive systems are defined by diversity and heterogeneity which entails complex relations and events that are influenced by actors (Rotmans & Loorbach, 2009). Narratives from actors of such a system can provide insights into how they practice within the current state of the systems and think of innovations or practices beyond existing regime to foster systems transitions (Geels, 2002). Narrative analysis is used to uncover the meanings encoded in the "narratives" or stories that interviewees tell (Kvale & Brinkmann, 2009). They go beyond depiction of reality, by

suggesting how reality is constructed through framing of linguistic patterns that communicate one's views (meanings, interpretations, and opinions) around a phenomenon. As no single narrative exclusively defines a particular discourse, no narrative is confined to just one discourse, as they can support multiple discourses despite varying interpretations by those who encounters them (Hermwille, 2016). Instead of just linking the events/ episodes, narratives can help construct a clear picture by weaving the distinct episodes (Alvesson & Sköldberg, 2000).

Through narratives, actors narrate challenges, and situate it in context of experiences, and/or efforts to resolve them. Kaplan (1993) views narratives as plotlines which constitutes a beginning, middle and an end (Kaplan, 1993). They begin with presenting an equilibrium (or existing state of affairs), followed by an event disrupting this equilibrium, and end with a new equilibrium (new state of affairs), thus presenting a temporal transition, or a change over time from one state to another (Ochs, 1997). A plot explains the reason for transition by defining the characters with relevant roles (protagonists and antagonists), assigning functions to individual events, and connecting the events into a meaningful sequence to form an interpretive theme (Czarniawska, 2000), p.10).

Accordingly, in Paper II, narrative analysis was conducted for transcribed interview material to develop plotlines as suggested by Ochs (1997). The data was read by at least two researchers several times, identifying the transition plotlines and developing the beginning (equilibrium), middle (event), and the end (new equilibrium, the roles of actors (protagonists and antagonists) (Kaplan, 1993; Ochs, 1997). Interpretive themes were also tested repeatedly until all the transcribed material was analyzed, and then were gathered under the three most relevant interpretive themes. The framework shown in Table 2 was inspired by Nordensvard and Ketola (2022) as used as an outline for the analysis.

Table 2. Narrative frame and the role of the policy storyteller (inspired by Nordensvard and Ketola (2022)) (from Paper I)

	Transition narrative frame	Policy storyteller
Characters	Pitting a protagonist and antagonist against each other, e.g.: Innovators vs. vested interest Dominant technology vs. niche innovation Status quo vs. change Public sector vs. private sector	The policy storyteller tells how and why this is a problem
Plotlines		
EQ(A): Where is the first state of equilibrium in the story?	The current situation and the protagonists' interests are compared and understood to be a conflict of interests and transition statement	The policy storyteller tells: What will happen if change does not occur?
Change: What disrupts EQ(A) and creates change?	The protagonists' suggestions for the solutions to the conflict	The policy teller tells: What is the solution? How will change happen? Who will lead this? Who will be a hinderance to the change?
EQ(B): Where is the next/last state of equilibrium in the plot?	The protagonists change the overall system by achieving change and challenging the status quo	The policy storyteller tells: the ideal outcome and what this transition leads to, and why this is desirable

4.3.4 Thematic analysis

Within Papers II and III, the thematic analysis strategy involved a six-step process as recommended by (Braun & Clarke, 2006). The first step involved reading the articles and becoming familiar with the content. The content was then coded inductively with relevance to the research questions. After repeated reading, the codes that emerged were identified and rechecked to see if they fit

better under other codes or if they needed to be discarded. Codes were compared with each other for similarity. This enabled analysing the codes for possible clusters with shared patterns of meaning (Braun & Clarke, 2006; Durach et al., 2017), and themes thus emerged. These themes were vehicles, airspace, vertiport, data links, energy systems, multimodal transportation, regional planning, traffic management, pilot and operator, environment, market, and society. These themes were then reviewed and defined by clarifying their scope. Further synthesising them provided aggregated themes, namely, primary technology, functionality, and adoption, forming a conceptual framework for the drone system. The thematic analysis of Paper II revealed potential trade-offs, which were outside the scope of the paper. These were noted down and further pursued in Paper III. For Paper III a similar inductive thematic analysis approach was taken which led to themes, namely, "flight altitude, airspace design, restricted zones, landing infrastructure, digital infrastructure, charging infrastructure, time of operations, type of delivery, transportation model, type of application, and choice of drone".

4.3.5 Quantitative assessments

In Papers IV and V of this thesis, quantitative assessments have been conducted to evaluate the energy consumption, and therein energy impact (in Paper IV and V) and emission impact (in Paper IV) of drone use as a delivery vehicle. The system boundary is defined as a last-mile delivery system for a specified delivery scenario, where selected transport vehicles perform the function of delivering packages to delivery stops. For emerging technologies, an ex-ante approach can be used for prospective analysis, i.e., to forecast the potential impacts of future technological systems (Cucurachi et al., 2018) approximately when the technological readiness and implementation is expected. Such ex-ante assumptions were therefore made about both drone development, and road vehicles. First is the choice of road vehicle to be compared against (like E-van in paper IV and electric truck in Paper V), choice of fuel in the IC-van in Paper IV (Hydrogenated vegetable oil or HVO, for primary calculations, and diesel for sensitivity analysis) and cost parameters used in Paper IV which are assumed based on predicted values for the years 2027-2032.

Energy and emission impact

In both the papers, a common approach was taken to evaluate drone energy impact in a delivery scenario. Considering a delivery scenario was the first step. Scenarios developed in paper IV was inspired by a real demonstration project at the co-creating company of this PhD project (SESAR, 2023) with drone delivery between Norrköping and Linköping. Scenario modelled in Paper V was based on lessons learnt from Paper IV and inputs from logistics companies.

The second step was to model the scenario. Location was considered which were geographical in Paper IV in urban area of Norrköping and in rural areas between Norrköping and Linköping in Sweden. The problem framing for Paper V was based on Ydre municipality in Sweden in Paper V which however was modelled more conceptually without being limited to a particular location.

Further, number of stops, payloads to be delivered at each stop, delivery distances (road distance for trucks, and point-to-point distance for drones), and speeds at which vehicles will travel based on road and vehicle constraints were determined. Vehicle parameters were then retrieved from manufacturer specifications, and literature reviews. In Paper V, the truck-drone hybrid configuration was also included in the scenario modelling.

The third step was to calculate energy use of the vehicles for the delivery scenario, where distance and payloads vary with stops. Energy use for drones was calculated using power equations for drones to overcome forces acting during different stages of flight, namely take-off or climb, hover-flight or cruise, landing or descend, and hover, and time taken at each stage, between different stops. Large drone was considered in Paper IV and thus energy equations from Kasliwal et al. (2019) were considered most suitable as the model developed in this paper was for a large drone. Similarly, small-drone was considered in Paper V due to which energy equations from Kirschstein (2020) as they make for a smaller drone. Energy and fuel use for vans were calculated using manufacturer specifications on energy or fuel consumption per km in Paper IV. In paper V, power equations for truck to overcome the forces acting on it, namely, rolling resistance, air drag, grade, and inertia are used from Kirschstein (2020); Kirschstein (2021); Kirschstein and Meisel (2015).

Finally, CO2 emissions were further calculated based on energy used by using emission factors (gCO2eq/kWh) for electricity and fuel, while Paper V evaluates energy reductions. The results for different vehicles considered in the respective study are compared and interpreted.

Economic impact

In Paper IV, economic impact assessment was conducted using life-cycle costing (LCC). LCC is a method used to assess the total cost of a product, system, or technology over its entire life span including initial investment, operation, maintenance, and disposal costs (Woodward, 1997). It helps organizations in making informed decisions about long-term investments, especially when evaluating emerging technologies. Initial costs, service and maintenance cost, operating costs, disposal costs, and vehicle insurance costs were included in the study based on ex-ante projections. Net present value equation was used to calculate the cost benefits for the years under consideration for ex-ante assumption (5 years).

Complete details on scenario development, data collection, formulae, and calculation for delivery time assessment, are presented in the appended papers of this thesis.

4.3 Reflections on research quality

The foundational criteria in research methods that guides rigor and quality in both qualitative and quantitative research. In qualitative research, this entails four trustworthiness criteria. Credibility or internal validity refers to the ensuring that data has been understood correctly with regards to the social world (Bryman, 2016; Kvale & Brinkmann, 2009). Transferability or external

validity regards to the generalizability of the findings in broader contexts (Bryman, 2016; Yin, 2009). Reliability refers to the consistency and replicability of the results such that results are reproducible under similar conditions (Bryman, 2016). Finally, confirmability which refers to ensuring objectivity of the researcher to avoid bias (Bryman, 2016). For quantitative research, such concepts of reliability and validity (both internal and external validity) are critical to maintain the quality of research (Bryman, 2016).

To ensure reliable research methods, preestablished calculations, and methodologies have been used for conducting literature reviews, interviews, energy and emissions calculations, and life-cycle costs (Papers I-V). Previous narrative frameworks have been used as inspiration to systematically guide narrative analysis (Paper I). These methodologies have also been systematically and transparently reported in the papers and in this thesis. Also, during literature review, search strategies and inclusion/exclusion criteria relevant to the research questions of the studies have been transparently reported. In SLR, only peer reviewed material was included which improved rigor.

In ILR, inclusion of grey literature could risk biases that may be inherent in documents like whitepapers (Garousi et al., 2019). However, the incorporation of these white papers was seen as necessary to strengthen the empirical evidence in an immature field of study. Additionally, the interviews served to triangulate results obtained from literature review (in Paper II and III) providing evidence for ensuring consistency and accuracy of results thus improving validity (Yin, 2009). In Paper V, inputs from logistics companies were used to formulate the study, thus improving reliability. The values assumed in quantitative studies were based on literature reviews (in Paper IV and V) as well as interviews from drone manufacturers which improves validity of the findings. Additionally, in quantitative papers (Paper IV and V) units have been defined and consistent throughout the papers and provides scenario models that are replicable. Two transcribers worked together to ensure accuracy of transcription. During narrative analysis (in Paper I), peers and supervisors were involved in interpreting the meaning of the narratives to ensure credibility. Narratives have also been described in a contextual manner through use of frames in Paper I.

One challenge with interviews is social-desirability bias. This type of bias can involve respondents who may have a tendency to present differently from who they are, either due to need for approval or to provide a socially desirable outcome such as overclaiming environmentally friendly behaviour (Brace, 2018). This was however minimized by guaranteeing confidentiality, keeping the questions simple, and asking follow-up questions (Bergen & Labonté, 2020; Brace, 2018; Kvale & Brinkmann, 2009)

5. Results

In this section, summaries of the five appended papers along with their key findings are presented.

5.1 Paper I

Regulatory landscape being revealed as one of the major barriers for drone implementation both in literature, in co-creative problem framing, and lack of research on the topic formed the basis to explore policy landscape for drone transportation in Sweden. Particular attention was given to explore policy narratives, which included how actors perceived the technology's potential to contribute or not to sustainable transport transitions; how they understood their own roles or roles of other actors in its implementation; the interplay between different actors to work with drone policy; how drone initiative were guided, prioritized, and managed; and how actors legitimized or delegitimized drones. Hence, this paper pursued two research questions: 1) How do actors narrate drone innovation and the roles for drone policy development in Sweden? and 2) How do actor narratives suggest the presence or absence of niche leadership to manage drone innovation in Sweden?

MLP was used, where electric drone transportation was located at the niche level where alterative and innovative practices and process take place, and relevant policy actors were situated on the regime level. Using MLP, this paper aimed to explore the narratives of relevant policy actors regarding their views on the role of drone innovation, roles of different actors, and their perspectives on transition using innovations like drone transportation for its future in sustainable energy transitions for transportation. To answer this, semi-structured interviews were used for data collection. The study employed a narrative analysis, focusing on different actors' narratives who are involved or affected by this transition, recognizing narratives as a form of discourse that constructs versions of reality. The analysis was conducted by focusing on the social meaning embedded in policy and practice to gain deeper understanding of current state of drone policy practices.

5.1.1 Narratives

The paper identified that the governance of drone transport is dominated by three narrative frames (as seen in table 3) market frame- where private actors are seen to lead drone transition, 2) collaboration frame- where public actors co-learn with private actors, but public actors are seen as important for drone implementation, and 3) non-priority frame- where actors do not prioritize innovations beyond the existing regimes.

Narrative 1 positions private actors, namely, startups, aviation firms, and technology developers, as the primary drivers of drone innovation. In this

frame, technological development and market forces are expected to lead integration, while public authorities remain largely reactive. Narrative 2 emphasizes joint leadership between public and private actors. This frame highlights the need for public authorities, particularly municipalities, regions, and airports, to take an active role in influencing regulation and infrastructure, while co-learning with industry actors who possess technical know-how. In both narrative 1 and 2, belief in drones serving societal function and need to support business interests can be seen. Narrative 3 frames drones as unnecessary or incompatible with sustainability goals, often associating them with conventional aviation and high emissions. Actors adopting this view prioritize established transport systems, particularly rail, and see drones as a distraction from climate objectives.

Table 3. Identified policy narrative frames on drone transportation (from Paper I)

	Market frame – private actors lead drone implementation	**Collaboration frame – public actors co-learn with private actors to lead drone implementation**	**Non-priority frame – public actors' priority is limited to pre-existing technologies**
Characters	Protagonist: technology, innovators, aviation industry, startup companies Antagonist: hackers, security issues Side characters: society, acceptance	Protagonist: private actors, public actors Antagonist: trains, politicians that want to close down airports, public organizations that don't cooperate Side characters: society, lack of knowledge	Protagonist: trains Antagonist: drones and aviation Side characters: flight shame
Plotlines			
EQ(A): Where is the first state of equilibrium in the story?	Using drone technology is in the nascent stage due to ecosystem development (supporting technologies for infrastructure)	Drone technology and its ecosystem is already developed and can contribute to society and business, but lacks policy support.	Drones are only relevant for delivery of medicine or extinguishing forest fires, not delivering goods or transporting people. We already have road

	Worst case scenario – drones will not get a chance to safely contribute to issues with ground transportation and business.	Worst case scenario – Drone transport won't be available for everyone if only private actors control it, only wealthy people will have access and be potentially dangerous to citizens and society.	transport that is more suitable. Worst case scenario: Consumers will continue travelling with aviation and we will not achieve our environmental goals.
Change: What disrupts EQ(A) and creates change?	The private actors, aviation companies and startup companies will lead the way and present new drone technology.	Private and public sector will collaborate and present regulations and infrastructure that will make the transition easier.	Climate change will mean moving away from aviation towards trains. We need to focus our resources on building fast trains
EQ(B): Where is the next/last state of equilibrium in the plot?	Drones will be a solution to transport goods and people, bringing society closer, and contribute to business interest. The technology and private sector will lead the way; the rest will follow and sort itself out	A safe space is created with clear regulations and infrastructure that will make drones useful and contribute to society and business interests. Public actors learn from private actors to then support them, while private actors enable public actors to take the lead	Society will use fast trains instead of aviation, which will lead to a more sustainable way of living

The paper finds these narratives to be contributing to a lack of leadership for drone policy development. Responsibilities for drone policy are seen to be fragmented, and many actors defer leadership to others. This is seen to reinforce polarization through narratives that are in disagreement, regarding the role of emerging niche like drones as well as understandings of sustainability in transport transitions. The paper thus identifies discursive lock-ins, that can hinder coordination, and risks entrenching existing transport regimes and narratives on sustainability transitions, rather than mature conversation, and coordinated efforts to experimentation with new socio-technical configurations specifically towards transformative ways to support Sweden's transport goals.

Such discursive lock-ins shows ideological conflict which plays a key role in decision making and thus influences niche innovation development. Creation of protective and regulated spaces for development and testing of innovations like drones are thus said to need nuanced discourses and mature conversations between actors with different ideologies and values.

The paper also emphasises the convergence of aviation and transportation regimes with drone innovation, where the development of the technology and systems currently remains within the aviation sectors, while the conventional transportation regime actors will act as customers procuring these services.

However, at the current stage, it is seen that the transport actors remain largely out of the discussion. Regulations of conventional transport logistics are another major barrier that drones will face if the intended application is last-mile deliveries. Thus, the lack of policy coordination, niche leadership, and dialogue between public and private actors is seen to pose as challenges for drone deployment to meet societal needs. Finally, the paper shows the need for new mental and discursive maps to navigate the complex policy landscape using MLP, which illustrates chaos, organizational silos, ontology and ideological divides, and the absence of mature conversations between various actors.

5.2 Paper II

Continuing with challenges with regulations for drone transportation systems identified through co-creative problem framing, particular lack of scientific knowledge associated with policy from a sustainability perspective, the research focus of exploration of policy needs for drone transport system was conducted. Thus, this paper addressed the drone transportation system through a regulatory lens to critically address the potential risks associated with implementing drones. The paper aimed to develop a policy framework for regional drone transport deployment through a systematic review of publications and interviews with key stakeholders, and hence pursued the following research questions: 1) What is the existing evidence base regarding policy requirements for drone transport system?; 2) What conceptual elements and detailed activities need to be considered in developing policies for drone transport system?; and 3) How can the policy requirements, conceptual elements, and detailed activities be synthesized into a policy framework for regional drone transport deployment?

These research questions were answered using systematic literature review, thematic analysis, and validated using interviews with experts from drone industry and relevant regulatory bodies, predominantly from a Swedish perspective. Outcomes from were further analysed in relation to EU regulatory frameworks for drones present at the time of this study, which included commission delegated regulation (EU) 2020/1058 (EU, 2020), commission implementing regulations (EU) 2021/664 (EU, 2021a), 2021/665 (EU, 2021b), 2021/666 (EU, 2021c), and 2022/425 (EU, 2022a), as well as EU Drone strategy 2.0 (EU, 2022b).

Paper II identified risks and policy needs related to drone transport system, which was categorised into themes, such as: 1) primary technology, which entailed vehicle-related aspects; 2) functionality, which entailed infrastructure and operations; and 3) adoption, which entailed the environment, market, and society (see figure 6).

5.2.1 Policy needs

Identified policy needs for drone technology included system-level and vehicle-level safety, vehicle activity, vehicle certification, and the reliability of autonomous systems.

Airspace-related policy needs identified were altitude regulation, navigation safety, airspace segregation and zoning, and integration with existing built infrastructure. Vertiports required technical standards and requirements as well as standards governing their locations. Policy needs for data links involve data-sharing standards and requirements, handling of system malfunctions, and issues of cybersecurity and privacy. Other infrastructure-related policy needs that were least discussed in literature were those concerning energy systems, regional planning, and multimodal transport. These include the integration of energy infrastructure into vertiport design, certification and authorization of charging or refuelling networks, energy storage technologies, and the standardization of battery swapping. In addition, policies are needed for zoning and land-use, regional airspace management, strategic positioning of drones in alignment with regional transportation goals, and integration with an efficient transition in existing transportation systems.

Traffic management policy needs relate to operational requirements, system architecture, safety, efficiency, maintenance, interactions between drones and manned aviation, and information sharing. Pilot and operator policies address licensing, certification, training, defined roles and responsibilities, and redundancy in the event of autonomous system failure.

For adoption, environmental policy needs include noise impacts on people and ecosystems, energy use and emissions, and visual impacts. Market-related policy needs involve pricing strategies, insurance, safety and service costs, and addressing the monopolization of airspace. Finally, societal policy needs concern job losses, impacts on local businesses, affordability and equity.

5.2.2 Policy framework

The paper showed the complexities involved in integrating drone transport into existing transport system that could exacerbate environmental issues or social inequalities if not assessed for local contexts. While top-down EASA regulatory frameworks are seen to address many of the technical aspects of the vehicle and its operations, many other aspects were seen to be under the jurisdiction of local actors. The complexity of governing drone systems was seen to need the establishment of organized institutional regimes for effective governance and enabling the involvement of local actors and citizens in the decision-making process so as to explore implementation based on local context and in line with sustainability use cases. The establishment of clear roles and responsibilities of various actors that are to be involved from a nation to local level was deemed necessary. Thus, policy framework as seen in Figure 6 is developed.

While validating with interviews, discrepancies are also seen between knowledge regarding the utilization of drones between public and private bodies, which have complementary and actor-specific knowledge. But knowledge of drone transport was seen to be low in local bodies. Therefore, coordinated and collaborative approaches through public-private co-creation is seen as crucial to combine technical expertise with that of regulatory oversight for adequately managing systems effects towards local needs. Capacity building of local and regional actors is also deemed necessary to manage the questions of impacts, infrastructure ownership and management effectively. Along this

line, for actors that want to explore this new mode of transport, regulatory sandboxes such as test beds and demonstrations are suggested to provide safe spaces to experiment, understand practical implications, developing the vehicles and infrastructure, and testing various designs and policies, to enable influencing the development of drones. Finally, the paper also advocates for setting up carbon budgets per Paris Agreement to steer the development of emerging technologies like drones to stay within the global emission reduction targets.

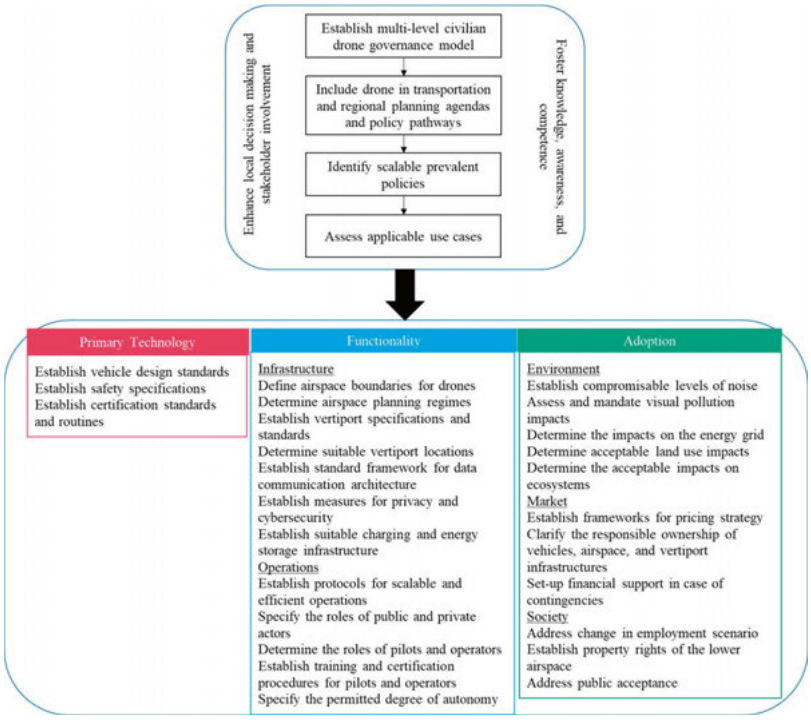


Figure 6. Policy framework and policy needs for civilian drone transport system (adapted from Paper II)

5.3 Paper III

Inclusion of local actors, particularly regional planners (from Paper II), transport actors (from Paper I) called for research addressing practical planning aspects when contemplating drone implementation. Due to the risks that drones would bring, public acceptance being another major consideration that was seen as necessary in Paper II in influencing how such planning decisions would be made. Public needs around drones are implicitly sustainability related aspects such as noise, safety, privacy, security, visual pollution, environmental issues, costs, and property rights. The interplay between planning considerations and public needs presents practical trade-offs that decision-makers will have to deal with. Therefore, this paper aimed to explore the dynamics between

have to deal with. Therefore, this paper aimed to explore the dynamics between such technological and infrastructural planning considerations and public needs in the context of integrating drone systems into existing transport networks. Two research questions were thus asked: 1) What are the key technological and infrastructural planning considerations for integrating drone systems into existing transport networks? and 2) What are the key implications of the dynamics between the technological and infrastructural considerations and public acceptance for regional and transportation planning?

To answer these research questions, an integrative literature review was conducted, validated with interviews. The literature review involved collection and analysis of both scientific and grey literature in the field that addressed technological and infrastructural planning aspects for drones. The interviews conducted with drone actors, relevant policy actors, and researchers, and served as a means to triangulate the results. The results were further critically analysed using the drone transport system map achieved from Paper II and a review of public acceptance factors, and a model as seen in Figure 7 was developed.

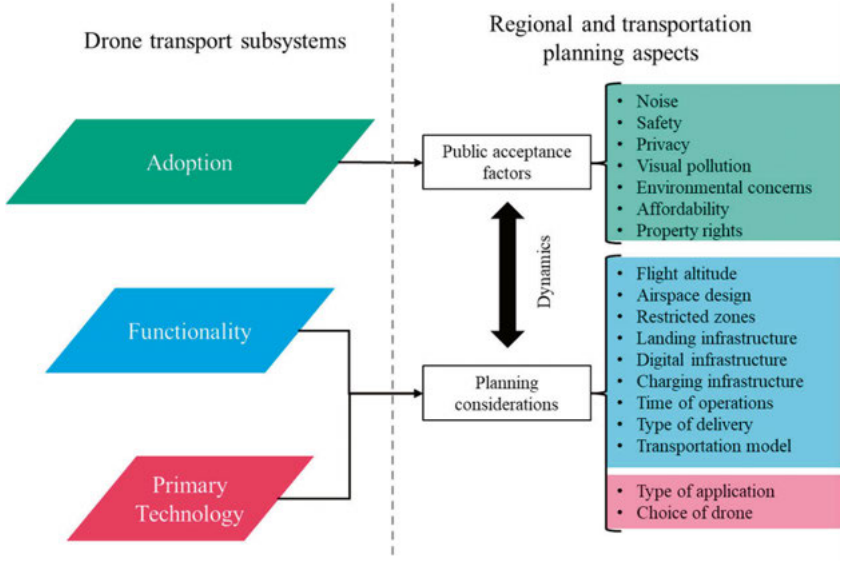


Figure 7. Identified planning considerations and their dynamics while planning with regards to public acceptance factors (from Paper III)

5.3.1 Planning considerations

This study identified planning considerations to be 1) type of application, 2) choice of drone, 3) flight altitudes, 4) airspace planning, 5) restricted zones, 6) vertiport locations, 7) digital infrastructure, 8) charging infrastructure, 9) time of operations, 10) type of delivery, and 11) transportation model; brief description will follow.

The unique requirements and impacts of various applications of drones, such as goods delivery, passenger transport (air taxis), and emergency services (medical equipment delivery, police surveillance, etc) affect planning from the initial stages of implementation to upscaled systems. Drones will be chosen for different applications based on their varying capabilities, such as payload capacity, range, and speed. Considerations towards the specific needs of the application to choose the most suitable drone type. Determining appropriate flight altitudes is essential for safety and regulatory compliance. Different altitudes may be required based on the type of operation and the surrounding airspace, which can help avoid conflicts with manned aircraft and ensure safe operations. Effective airspace management is significant for integrating drones into existing built environment and air traffic systems. This includes designing airspace structures (e.g., layers, cells, zones) that accommodate drone operations while ensuring meeting public needs. Identifying areas where drone operations may be restricted or prohibited is necessary for privacy, operational requirements, and flexibility. These zones may include locations near airports, military bases, or densely populated urban areas, where drone flights could pose safety risks. Planning for vertiports, which are designated take-off and landing areas for drones may involve repurposing existing infrastructure or developing new sites to facilitate efficient drone operations

Robust digital infrastructure is required to support drone operations, including communication systems for real-time data exchange, monitoring capabilities, and operational management. This infrastructure needs to be assessed for ensuring the reliability and safety of drone services. In addition, establishing charging facilities for electric drones is critical in these landing areas. Defining operational hours for drone flights is important to address public concerns regarding noise and to comply with local regulations. Different delivery methods, such as door-step delivery, rooftop deliveries, or pick-up points may require distinct planning approaches. Understanding the implications of each method on logistics and service delivery is essential. Finally, developing a suitable transportation model that integrates drones with existing transport systems is crucial by considering how drones can complement traditional logistics and public transport systems to enhance overall connectivity and efficiency.

5.3.2 System dynamics and implications for planning

Based on planning considerations identified, the systemic effects of various decision-making aspects for each category were analyzed, developing eleven key implications for regional planners, transportation actors, and future research. The study highlights several interconnected implications to be considered while planning to implement drone transport. First, it emphasizes evaluating choice of drones and the role it will play for parameters such as noise, energy efficiency, payloads, and speed. Additionally, the utilization of lower airspace requires evaluation of its potential impacts on biodiversity, property rights, and ecosystem services, with a strong emphasis on adopting a citizen participation approach. Designing airspace systems will account for factors like noise levels, capacity, safety, privacy, and energy efficiency, to meet the specific needs of different regions. The implementation of fly and no-fly zones

should be guided by application-based and time-specific scheduling to regulate drone operations. Furthermore, optimizing flight altitude is crucial to strike a balance between energy efficiency, noise impacts, privacy concerns, and safety requirements, depending on the drones in use.

Paper III also shows the need to evaluate infrastructure needs of landing sites based on the type of drone services, connectivity, noise levels, service equality, and the addressing challenges posed by dispersed settlements. Ensuring adequate digital connectivity in areas where drones are expected to operate is also essential to support reliable communication and safety systems. The energy infrastructure for drone operations must be analyzed for its grid capacity, scalability, land use, battery capacity, number of charging stations, and the strategic placement of charging stations to enable sustainable and efficient operations. A critical assessment is also needed to weigh the trade-offs between enhancing accessibility through drone transport and the potential long-term environmental impacts. Moreover, the design of drone-related infrastructure should prioritize multifunctionality, resource efficiency, and relevant public concerns. Finally, there is a need to reflect on the broader, long-term systemic impacts of ownership (of drones and their infrastructure), the diversity of actors involved, and the evolution of transport models, to ensure that these developments contribute meaningfully to regional sustainability goals.

5.4 Paper IV

The lack of clarity on the impacts of drone transportation (Paper II) particularly in Sweden provided the rationale for conducting this quantitative study. To better understand the emissions and operational impact for logistics from an energy, costs, and time perspective, a study on impact assessment for electric drone for last-mile delivery was conducted. The aim of this paper was to assess CO2 emissions, life-cycle costs, and delivery time for delivery routes in urban and rural areas. A drone demonstration project between Linköping and Norrköping from practice (SESAR, 2023) served as a basis for developing cases for this study. Hence, the scenarios developed were a long and short route between Linköping and Norrköping (long route between cities: LBC and short route between cities: SBC), which primarily consisted of rural areas between the cities, and a long and short route were developed within the city limits of Norrköping (long route within cities: LWC and short route within cities: SWC). A comparison was made with E-vans and IC-vans for deliveries in the same routes. Please note that the ground vehicles were called “light-weight trucks” in Paper IV but will be more appropriately referred to as “vans” in this thesis. The functional unit chosen for the study was one standard delivery of 250 packages, weighing 2 kg each, divided into three stops, with maximum payload of 500 kg. A large drone with a payload of 500kg was chosen to get a fair assessment with the vans.

Paper IV consisted of 1) energy and emission assessment for the use phase of delivery vehicles, which included calculations of energy consumption using formulae for the different phases of drone flight (hover, climb, cruise, descend); 2) life-cycle cost assessment; and 3) delivery time. These calculations were

performed for all three vehicles and comparisons were made. Data was collected using government records, manufacturer website, and interviews. Furthermore, sensitivity analysis was conducted for varied emission factors for electricity and fuel, drone energy use, reduced drone flight distance, number of drone deliveries, and drone speed. Some of the most interesting results are mentioned below.

5.4.1 Energy and emissions impact

Paper IV showed large drone to have the highest relative energy use across the delivery scenarios with ex-ante assumptions (as shown in Figure 8). In contrast, E-van consistently exhibited the lowest energy consumption, using approximately 75% (LBC), 80% (SBC), 82% (LWC), and only 92% less energy than the drone in the SWC scenario (i.e., around 8% of drone energy use), which naturally remains the same for emissions (as seen in Figure 9). IC-van showed higher energy use than E-van but still demonstrates substantial energy savings relative to the drone, with reductions of about 12% (LBC), 31% (SBC), 40% (LWC), and 74% (SWC). However, due to fuel emissions, HVO fuelled IC-van can be seen to emit more than the large drone for longer routes, with increases of approximately 36% for LBC and 8% for SBC, while showing modest reductions for LWC (about 6%) and a higher reduction for SWC (around 59%). In contrast, while conducting sensitivity analysis with diesel emission factors, it was seen that diesel fuelled IC vans can have emission factors from one and a half time of the drone (for the shortest route SWC) to up to almost four times the drone (for the longest route LBC). Thus, the results indicate that E-vans have the lowest energy use and emissions impact. The large drone has higher energy use than IC-vans but exhibit significantly lesser emissions than diesel in the IC-van. With regard to HVO fuelled IC-vans, large drones perform better on longer routes, while HVO is a better alternative for shorter ones.

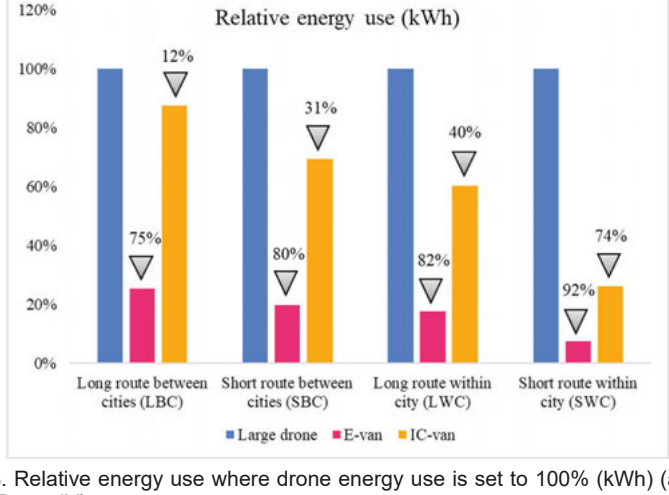


Figure 8. Relative energy use where drone energy use is set to 100% (kWh) (as calculated in Paper IV)

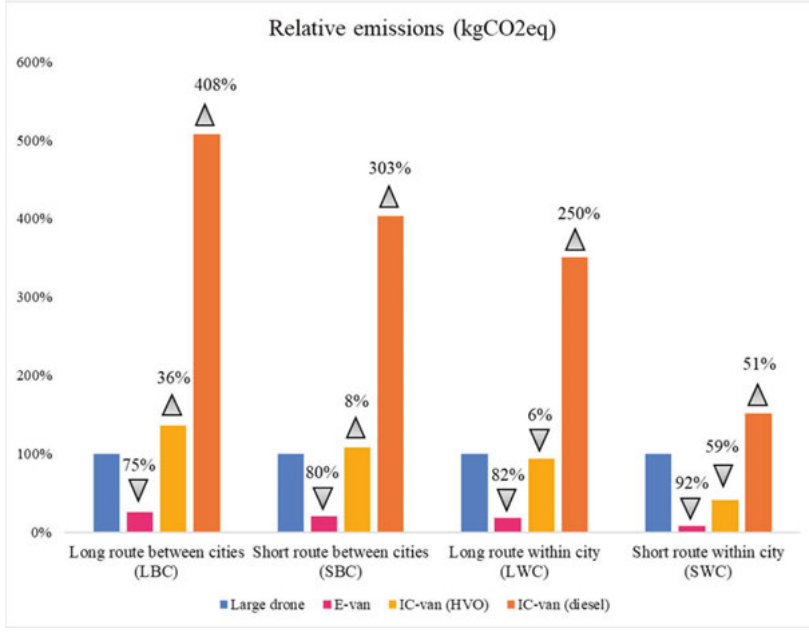


Figure 9. Relative CO2 emissions of vans where drone emissions is set to 100% (from Paper IV)

The sensitivity analysis was conducted by comparing CO2 emissions for varied drone energy demand and the distance it travels, to the emissions of the vans for the same distance. This provides indications on when the drone starts to be a feasible option as a lower emission alternative (shown in Figure 10). Based on these results, the drone energy demand required to consider replacing an alternative road vehicle is approximately ≤3-3.5 kWh/km for HVO fuelled IC-vans, and ≤0.5 kWh/km compared to e-vans, assuming moderate to high distance reductions. Thus, a conclusion was that a smaller drone which has significantly low energy demand and significantly lower range would be more likely to be an appropriate fit, particularly for short distance deliveries.

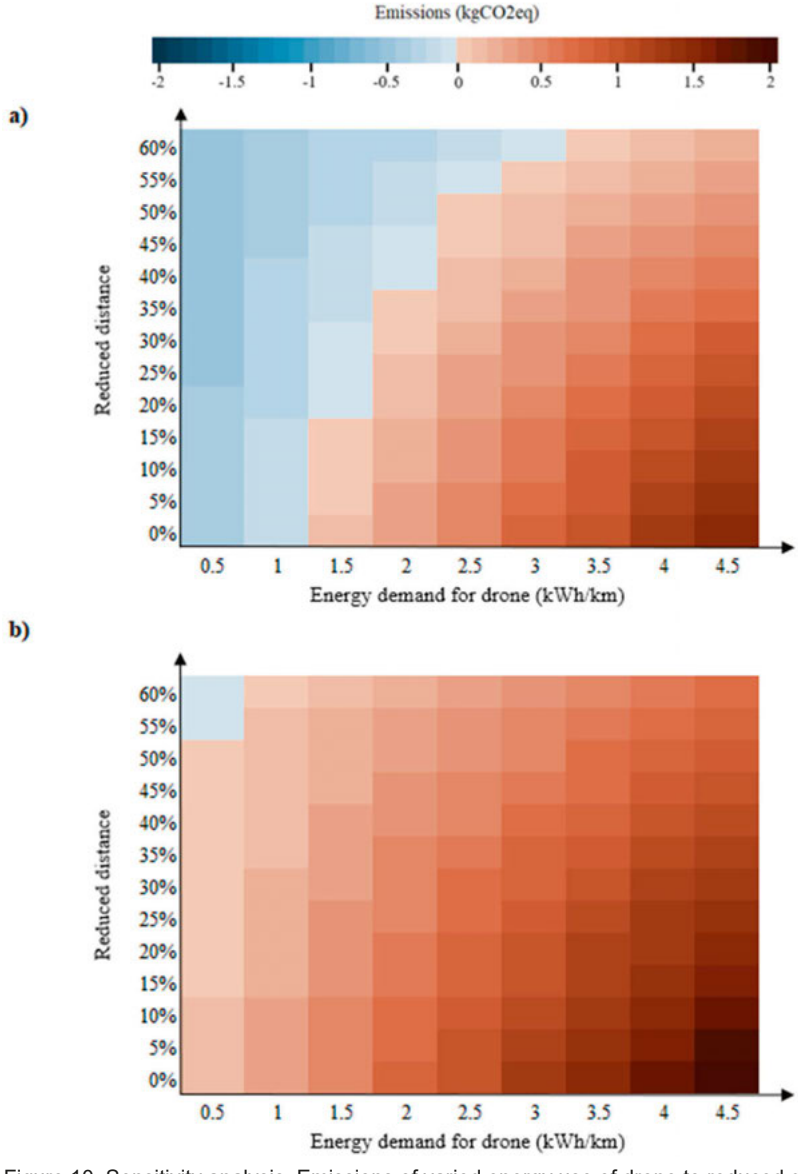


Figure 10. Sensitivity analysis: Emissions of varied energy use of drone to reduced distances for van and drone. a) compared to HVO fuelled IC-van, and b) compared to E-van (from Paper IV)

5.4.2 Economic impact

The life-cycle cost analysis showed large drone to exhibit the lowest costs across all route types (see Figure 11). Both E-vans and IC-vans exhibit substantially higher relative costs, ranging from about 117-165% for E-vans and 131-171% for IC-vans, depending on the route. The cost breakdown shown in

Figure 12 shows the main difference to be staff costs due to automation. For the vans, staff costs are the highest, accounting for roughly 70-76% of total costs. Hence, automation compensates high initial costs of the drone, its maintenance, insurance, etc. Naturally, during sensitivity analysis, increasing drone operation to 24 hrs from 8 hrs further reduced costs significantly.

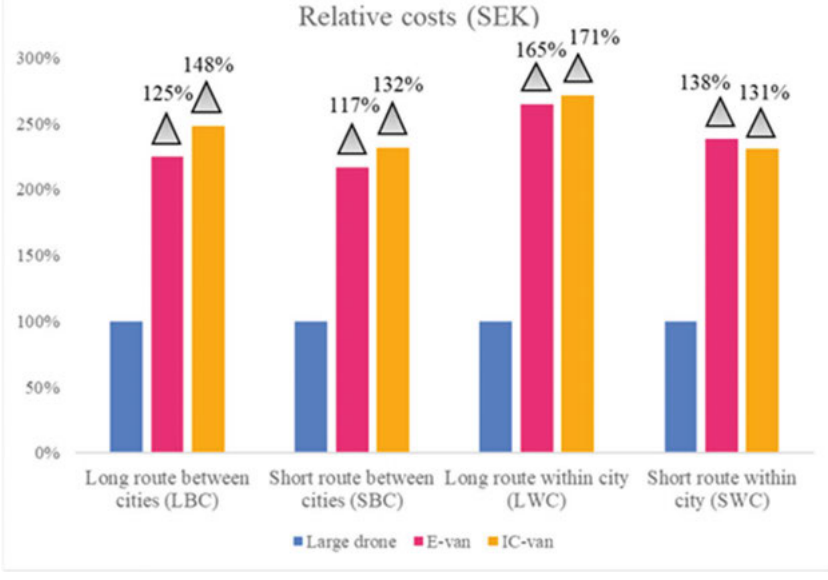


Figure 11. Relative life-cycle costs of vans where drone life-cycle cost is set to 100% (from Paper IV)

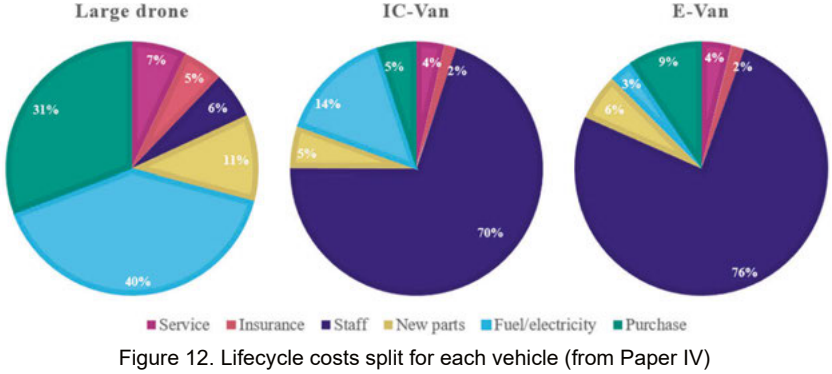


Figure 12. Lifecycle costs split for each vehicle (from Paper IV)

5.4.3 Delivery time

The delivery time comparison showed that drones were consistently the fastest option across all route types (see Figure 13). Van-based deliveries require substantially more time, with relative delivery times of about 142% for long routes between cities, 128% for short routes between cities, 165% for long routes within cities, and 136% for short routes within cities. The results highlight

delivery speed as a key operational advantage of drone-based delivery compared with van-based alternatives.

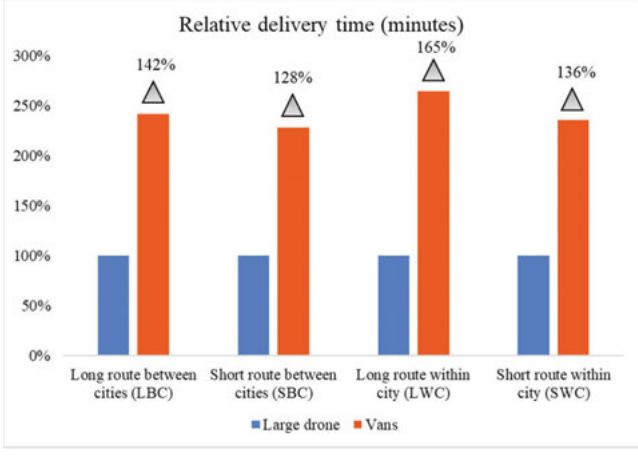


Figure 13. Relative delivery time of vans where drone delivery time is set to 100% (from Paper IV)

5.5 Paper V

The results from Paper IV provided the motivation to explore energy use of multimodal truck-drone logistics where smaller drones complement the tuck by conducting deliveries cooperatively. Additionally, need for including logistics companies (as seen in Paper I and III) were deemed crucial to address real logistics problems such as addressing truck loading and infeasibility of truck-based delivery for low-density areas. This provided the rationale for developing an adaptable model was chosen to include delivery density (number of stops and distance of service area) and truck payload as the operational parameters to be tested. This paper pursued quantitative evidence on when and under what conditions drones can contribute positively to energy-efficient transport by focussing on the operational energy performance of truck-drone hybrid delivery systems in last-mile logistics.

The aim of Paper V was thus to evaluate how integrating drones with delivery trucks impacts energy use in last-mile delivery, focussing on operational parameters, namely, truck payload, number of deliveries stops, and service area radius. It further sought to identify the operational conditions under which such hybrid systems achieve optimal energy performance. The analysis was conducted for both IC-truck and E-truck. Based on this aim, the paper addressed two research questions: 1) How does the integration of drones with delivery trucks affect the total energy use of last-mile delivery systems? 2) How do key operational parameters, namely truck payload, number of delivery stops, and service area radius influence the energy efficiency of hybrid truck-drone delivery systems to provide optimal performance?

To answer these questions, Paper V developed a simulation model representing last-mile delivery operations, using randomized delivery locations and mixed payload distribution. One truck equipped with two electric drones was modelled, where drones operated as add-ons launching from and returning to the truck along its route. Truck routes were optimized using a travelling salesman function, while drones were assigned only to deliveries that were within their payload, and range. Battery swapping is assumed to be done by the truck driver, and around 2% of the truck load is dedicated to batteries. Sensitivity analyses were performed by systematically varying truck payload levels (25-100%), number of delivery stops (20-150), and service area radius (5-15 km). For each configuration, 1,000 simulation iterations were conducted, and operational energy use was calculated for both truck-only and truck-drone delivery systems.

5.5.1 Energy impact

The results show that truck-drone hybridization consistently reduced operational energy use compared with truck-only delivery across all tested parameters. Energy reduction was observed for both IC and electric trucks under all truck payload levels, service radii, and delivery stops. For varied number of delivery stops (shown in table 6), energy reduction peaks at moderate numbers of stops (approx. 30-70) and gradually declined as the number of stops increased, though hybrid systems continued to outperform truck-only operations even at high stop densities. Shorter service radii were seen to increase the benefits of hybridization, with the largest energy reductions occurring at a 5 km radius (as shown in table 5). Truck payload level had a comparatively smaller influence on hybrid-system energy performance, with relatively minor variation in energy reduction for decreased payload levels (see table 4). While drone assistance for IC and E-trucks demonstrated consistent and significant efficiency gains, it was observed that IC truck- drone systems achieved slightly higher relative energy reductions than electric truck-drone systems.

Multiple parameters were further varied to check their influence on energy reductions and study the patterns. The results showed the spatial and operational density variables, namely service radius and delivery stops to have most influence on energy reductions, while truck payload plays a secondary but still relevant role.

In tables 7 and 8, energy reductions are seen to rapidly increase as the number of delivery stops rises, reach a maximum, and gradually diminishes as number of stops further increase. For IC truck-drone systems, peak reductions occur between 30-60 stops when seen in line with payload variation and 30-50 stops when seen in line with varying service radius. For E truck-drone systems the peak is seen between 40-60 stops when seen in line with payload variation and 40-60 stops when seen in line with varying service radius. Beyond these ranges of delivery stops, the energy reductions for both trucks start to decline, but are still seen to remain significant. Payload effects are most pronounced at low stop counts where energy reductions are substantially higher for low payload than at full payload, especially for routes with only 20-30 stops (see table 1). In table 9, it can be observed that in service area with short radii (5 km),

hybrid systems show largest reductions of about 21-22% for E truck-drone, and around 23% for IC truck-drone delivery.

Table 4. Energy use effect of truck-drone delivery across varying truck payloads (from Paper V)

Truck payload variation for 65 stops, 10 km service radius	**IC truck-drone Energy use effect (%)**	**E truck-drone Energy use effect (%)**
100%	-15,0%	-14,1%
75%	-14,9%	-13,9%
50%	-14,7%	-13,4%
25%	-14,5%	-13,0%

Table 5. Energy use effect of truck-drone delivery across varying service radius (from Paper V)

Service radius variation (km) for 65 stops, 50% truck payload	**IC truck-drone Energy use effect (%)**	**E truck-drone Energy use effect (%)**
15	-11,6%	-10,5%
10	-14,8%	-13,5%
5	-22,9%	-21,7%

This indicates that when trucks operate under partial loading conditions and for sparse delivery densities, drones can contribute more effectively to overall energy savings. However, as stop density increases beyond about 70 stops, the influence of payload diminishes and energy savings converge across payload levels, showing that payload becomes largely irrelevant at high stop densities.

This paper shows how smaller drones can provide more flexibility for truck deliveries in the last mile. By reallocating feasible tasks to drones, trucks get opportunity to further optimize their routes. Thus, small drones are seen to complement both IC and E-trucks in create energy-efficient multi-modal system. This is seen as particularly relevant when using trucks for higher number of stops number of stops. It can also be concluded that such a system can have significant reduction in both high density and low-density areas. Most inefficient to use drones in delivery scenarios with very low number of stops especially when the trucks are heavily loaded as it does not have enough stops that are allocatable to drones. Optimally, hybrid drone-truck deliveries are seen to achieve energy savings in the range of around 13-23%, with optimal ranges being 40-50 stops, 5-10km service radius, and 25-50% payloads, however, these are highly dependent on the specific contexts of application.

Table 6. Energy use effect of truck-drone delivery across varying number of delivery stops (from Paper V)

Stop variation for 50% truck payload, 10 km service radius	**IC truck-drone Energy use effect (%)**	**E truck-drone Energy use effect (%)**
20	-8,1%	-4,8%
30	-16,7%	-11,9%
40	-17,2%	-13,7%
50	-16,2%	-13,9%
60	-15,0%	-13,5%
70	-14,2%	-13,2%
80	-13,3%	-12,7%
90	-12,6%	-12,2%
100	-11,9%	-11,7%
110	-11,5%	-11,4%
120	-10,9%	-10,9%
130	-10,4%	-10,5%
140	-10,0%	-10,1%
150	-9,8%	-9,9%

Table 7. Energy effects from truck-drone deliveries across varying number of stops and payload levels, at 10km service radius (from Paper V)

Truck pay-load	100%	75%	50%	25%	100%	75%	50%	25%
Number of stops	**IC truck-drone Energy use effect (%)**				**E truck-drone Energy use effect (%)**			
20	-1,7	-3,9	-7,7	-16,2	-1,1	-2,4	-4,5	-9,3
30	-8,2	-12,2	-16,7	-17,8	-6,0	-8,7	-11,9	-12,2
40	-13,7	-16,0	-17,2	-16,8	-11,1	-13,0	-13,7	-13,0
50	-15,3	-16,3	-16,2	-15,8	-13,4	-14,3	-13,9	-13,2
60	-15,2	-15,5	-15,0	-15,1	-14,1	-14,2	-13,5	-13,3
70	-14,5	-14,5	-14,2	-14,1	-13,9	-13,7	-13,2	-12,8
80	-14,0	-13,6	-13,3	-13,4	-13,7	-13,2	-12,7	-12,4
90	-13,4	-13,1	-12,6	-12,6	-13,3	-12,9	-12,2	-11,8
100	-12,6	-12,3	-11,9	-11,9	-12,6	-12,2	-11,7	-11,3
110	-12,1	-11,8	-11,5	-11,4	-12,3	-11,8	-11,4	-11,0
120	-11,4	-11,2	-10,9	-11,0	-11,7	-11,4	-10,9	-10,7
130	-10,9	-10,8	-10,4	-10,5	-11,2	-11,0	-10,5	-10,2
140	-10,6	-10,2	-10,0	-10,1	-10,9	-10,5	-10,1	-9,9
150	-10,2	-9,9	-9,8	-9,7	-10,6	-10,2	-9,9	-9,5

Table 8. Energy effects from truck-drone deliveries across varying number of stops and service radius, at 50% truck payload (from Paper V)

Service radius (km)	5	10	15	5	10	15
Number of stops	**IC truck-drone Energy use effect (%)**			**E truck-drone Energy use effect (%)**		
20	-8,4	-8,1	-7,1	-5,4	-4,8	-4,2
30	-21,1	-16,7	-15,4	-16,2	-11,9	-10,6
40	-25,5	-17,2	-15,2	-21,7	-13,7	-11,6
50	-24,8	-16,2	-13,8	-22,4	-13,9	-11,2
60	-23,5	-15,0	-12,6	-22,0	-13,5	-10,7
70	-22,2	-14,2	-11,5	-21,2	-13,2	-10,2
80	-21,1	-13,3	-10,7	-20,6	-12,7	-9,7
90	-20,1	-12,6	-10,0	-19,9	-12,2	-9,2
100	-18,9	-11,9	-9,3	-18,9	-11,7	-8,8
110	-18,2	-11,5	-8,9	-18,3	-11,4	-8,5
120	-17,4	-10,9	-8,4	-17,6	-10,9	-8,1
130	-16,9	-10,4	-8,0	-17,2	-10,5	-7,8
140	-15,9	-10,0	-7,7	-16,2	-10,1	-7,5
150	-15,4	-9,8	-7,4	-15,7	-9,9	-7,3

Table 9. Energy effects from truck-drone deliveries across varying truck payloads and service radius, at 65 stops (from Paper V)

Truck payload	100%	75%	50%	25%	100%	75%	50%	25%
Service radius (km)	**IC truck-drone Energy use effect (%)**				**E truck-drone Energy use effect (%)**			
5	-23,1	-23,2	-23,1	-22,9	-22,3	-22,3	-21,9	-21,3
10	-15,0	-14,9	-14,7	-14,5	-14,1	-13,9	-13,4	-13,0
15	-12,3	-12,4	-12,1	-11,9	-11,1	-11,0	-10,4	-10,0

6. Concluding discussion

In this section, the research questions of this thesis are discussed based on the findings of appended papers and their implications.

6.1 Impacts and Potential Roles of Civilian Drones in Sustainable Transport Transitions

RQ 1. What impacts and potential roles can civilian application of drones have in contributing to sustainability transitions within the transport sector?

In this thesis, the civilian application of drones is understood as an emerging technological niche with the potential to redesign transport systems in ways that may significantly influence ongoing sustainability transitions. The papers, appended in this thesis, reveal indications of drone effects that may inform their roles for societal transitions. Their potential contribution for transport transitions is multifaceted with energy and environmental, operational, and systemic impacts that are further discussed.

6.1.1 Roles based on energy and emissions impacts

The environmental impacts of drones is highly context dependent, as suggested by many studies published during the time of the thesis (Aster et al., 2025). Quantitative papers IV and V provide unique new insights into the emissions and energy performance of drone-based transport. Large drones demonstrate the ability to reduce emissions under conditions that allow them to take advantage of efficient cruise phases and avoid emission intensive constraints of IC-vans. For example, in Paper IV, large electric drone exhibits lower CO2 emissions compared to IC-van fuelled by HVO, with reductions of up to 36% in LBC and 8% in SBC. However, the drone's energy use is still 12% and 31% higher compared to HVO fuelled IC-van. However, compared to IC-van fuelled by diesel, the large drone is seen to have lesser emissions (albeit higher energy) in all scenarios. However, E-vans have the least energy and emissions impact. Similar conclusion is drawn in Yowtak et al. (2020) where life-cycle assessment is conducted for a smaller drone. While studies generally do show that the appropriate use of drones for environmental benefits per se is in the rural areas, they generally do consider nano and micro drones (from classification presented earlier in the thesis, see figure 1) to arrive at this conclusion (for e.g., Goodchild and Toy (2018) considers hypothetical drone energy requirements that vary between 10 to 100 Wh/mile, and Park et al. (2018) considers a microdrone with 1.2kg payload).

This provides clear indications for choice of E-van as the alternative vehicle with least energy and emissions impact. However, since the attention is on drones in this thesis, it can be concluded that, from a CO2 emissions perspective, the use of drones is appropriate when contemplating its choice over diesel alternative.

Use of small drones in truck-drone configurations has been more explored to be designed operationally (Baik et al., 2025; Baldisseri et al., 2022; Wang et al., 2024). In Paper V we can see that when smaller drones assist trucks in hybrid truck-drone delivery systems, the configuration consistently demonstrates energy savings, in optimal conditions between 13%- 23%. Such configurations enable trucks to have more opportunity for efficient routing by allocating dispersed stops at shorter distances to drones. Thus, in such truck-drone delivery, small drones provide flexibility to improve energy performance. This brings forth the complementary and adaptive nature of drones that can be used for reducing energy use within last-mile logistics operations.

As current strategies for transport transition primarily involve continuing use of diesel vehicles with more sustainable fuels, or electrification of vehicles, use of drones can be synergistic with this strategy for accelerating energy-efficient transition. Drone integration can be parallel and coevolve with the development and implementation of other transport alternatives, enabling them to contribute to system improvement through hybridization. Such role enables the technology to have a symbiotic or synergistic relationship with the regime and enter main-stream markets (Raven, 2007).

Besides such benefits, adverse effects of drones can be seen in paper IV, which indicates large drones to have negative impacts in dense urban environments. Other studies in the field show that in short-distance, high-frequency delivery settings, where drones must repeatedly take off and land, their energy use and corresponding emissions can be high (Kasliwal et al., 2019). This is supported by this thesis, where large drones are shown to be energy intensive. Sensitivity analysis in paper IV also shows that the drones need to have very low energy requirement of about 0.5 kWh/km, and very low distances of at least 55% lesser compared to that of the E-vans, and about 1.5kWh/km to be compared to HVO fuelled IC van. Such low energy consumption is possible only through much smaller drones. While the performance of large and small drones (in addition to variations to how they are designed) varies significantly and obviously cannot be put under the same generalizable umbrella, the outcome of paper IV shows energy-intensive nature of large drones, meanwhile, smaller drones too are seen to be highly energy-intensive when the unit of measurement is per-ton km (Rodrigues et al., 2022). Therefore, it is important to view drones as an add-on to the system that entails its own set of impacts.

6.1.2 Roles based on operational impact

In terms of costs, which has been preliminarily explored in Paper IV, drones exhibit potential to lower operational costs in freight and last-mile delivery services. Life-cycle cost analyses show that large drones operate at less than half the cost per functional unit compared to both IC and E-vans. Although their initial investment costs are higher than the van, their automation

(assuming 30 drones can be operated by 1 operator) reduces the staff costs in truck-based delivery systems. Other studies also show that drones reduce costs by up to 41% compared to cars and increases amount of same-day deliveries (Ramos & Vigo, 2023) as well as reduce costs by 96.5% compared to truck delivery (Mohamed & Mohamed, 2025). Apart from this, route and speed optimization increases chances for reducing energy costs (Dukkanci et al., 2020; Kim et al., 2023) which is however not evaluated in this thesis. The ability of drones to provide significantly faster deliveries than vans in all studied scenarios in Paper IV also indicates high operational value.

Faster and cost-effective suggests that drones can be interesting to businesses by being economically competitive, thus positioning drones as a potential tool for establishing low-cost logistics systems. Its advantages can be immediate in the context of use for healthcare and other emergency services that seek to improve public services with constrained budgets. During emergencies or extreme weather events where roads become inaccessible, or when logistics gets disrupted, drones can continue to operate. Several international examples such as medical drone deliveries in Rwanda, show that drones can provide life-saving services where ground transport is unreliable (Kremer et al., 2023). These applications create a niche for drones as enablers of cost-effective services.

6.1.3 Roles based on spatial and structural impacts

With drones being a new mode of transport, new spatial and structural impacts emerge, that ultimately impact how societal transitions occur. One primary characteristic of drones often considered advantageous is the lack of need for road infrastruc-ture. There is need, however, for other ground-based infrastructure prerequi-sites such as take-off and landing sites or vertiports, digital connectivity, charg-ing stations, and warehouse distribution depending on the operational model as identified in Paper II and III. Such dedicated infrastructure is suggested to lead to new forms of land consumption (Straubinger, 2019). However, drones can be used in operations or configurations that do not rely on fixed landing infrastructure and instead utilise existing built infrastructure like rooftops or open area delivery points (as mentioned in Paper III). In cases where new infrastructure is needed, reusing existing infrastructure, particularly mobility hubs or rooftops, can minimise land consumption and support integration with other transport modes. Paper III highlights uncertainty regarding future elec-tricity demand as electrification accelerates across the transport sector, raising concerns about whether existing regional grid capacities can support rapid electrification (Palm, 2021; Pillai et al., 2012). In addition, reliable digital con-nectivity is identified in both papers as a prerequisite for safe operations be-yond-visual-line-of-sight, meaning that areas with limited network infrastruc-ture may face barriers to adoption until sufficient digital capacity is estab-lished. However, interest in drone adoption could improve digital connectivity in regions.

Changes to the use of the lower airspace is another key dimension. Paper II and Paper III identify the need to design and regulate the lower airspace to ensure safe, efficient, and acceptable operations, with considerations such as

flight paths and no-fly zones being influenced by proximity to critical areas and residences. There is also a risk of monopolisation of lower airspace access (Straubinger, 2019), depending on how planning and property rights are regulated (Bauranov & Rakas, 2021), suggesting that access could become limited to actors with financial or technological capacity rather than treated as a shared public resource or commons.

Drones as an additional transport mode therefore introduces planning considerations that differ from current approaches used in both urban planning and aviation sectors. The results suggest a shift from planning focused on two-dimensional infrastructure toward planning that must integrate both horizontal and vertical dimensions, where airspace becomes planned, zoned, and regulated in relation to the built environment (Bauranov & Rakas, 2021) with citizen's rights specified (Dolata & Schwabe, 2023). Here, traditional jurisdictional boundaries between municipal planning and aviation authority responsibilities become less distinct, requiring new coordination mechanisms and decision-making.

Paper I suggests that airports may establish new business models involving drone operations, turning into multimodal logistics hubs, or serving as connection points in emerging aerial networks, which may reposition airports within regional mobility systems beyond traditional aviation roles. Paper III discusses how drones may support more point-to-point logistics and passenger flows, reducing reliance on conventional hub-and-spoke patterns and potentially altering how spatial connections are formed within regions.

Paper II and III indicates that increased accessibility enabled by drones could contribute to urban sprawl if people relocate further from service centres while maintaining fast access to goods and essential services (Rendón et al., 2021), though this remains uncertain. At the same time, the same mechanisms may provide positive effects in rural areas by improving connections where conventional transport options are limited or expensive to operate, aligning drones more closely with objectives to provide equal services.

Thus, drones introduce new layers of spatial design, infrastructure planning, and direct and indirect effects. Their potential benefits depend on how infrastructure is designed, used or reused, how the lower airspace is governed, how energy and digital systems evolve, etc. This shift can be considered transformative as it introduces new infrastructure types, governance questions, and operational requirements that are not, to a large extent, part of current planning frameworks and challenge existing planning routines. Spatially, drones also transform how accessibility is conceptualised by adding a new dimension of mobility. Planning decisions influence how space is allocated, how mobility networks are envisioned, what interests are prioritized, and how future development patterns could evolve.

6.2 Barriers and Drivers Influencing the Integration of Drones for Energy Efficient Transport Transition

RQ 2. What barriers and drivers influence the integration of drones for transport transition?

The integration of drones into transport systems requires co-evolution of digital and physical infrastructures, and policy landscape that enables safe drone adoption. Such developments require complex vertical and horizontal collaborations (Howlett & Del Rio, 2015). In this thesis, focus has been placed on understanding the regime dynamics involved in the integration of technology and its desirability for regime actors to integrate it to address systems lock-ins. Such aspects can be both barriers and drivers depending on the constructed understanding of the technology as well as its effects. Some factors identified in this thesis are addressed in this section. Although these drivers and barriers influence technology adoption, some of them are more important for purposive adoption of the technology intended to meet specific sustainability goals. Purposive transitions particularly require high coordination and external resources to meet certain sets of interest (Berkhout et al., 2004); in this thesis, these interests concern is the use of drone for transitions towards sustainable transport systems.

6.2.1 Narratives as barriers and drivers

Addressing carbon lock-ins can be seen as a discursive challenge that also entails integration of technological solutions as part of energy-efficient futures (Buschmann & Oels, 2019). Institutional practices are influenced by discourses that can either limit or support collaboration among multiple actors (Geels, 2002) which also affects in gathering interest of other actors to experiment with alternative technologies which dictates its development, gathering market interest, and become part of new regimes (Smith et al., 2005) that deviate from existing regimes and in turn deviate from institutional practices (Phillips et al., 2004).

Paper I shows that how stakeholders frame their value, allocate responsibility, negotiate legitimacy, and interpret sustainability are key areas of conflict and determine the future with or without technologies like drones. Positive discourse for drones creates strong legitimacy, interest, and thus a safe space for actors to become involved for experimenting with drones, to provide good preconditions and drive for drones to be adopted. Why drones are to be adopted is however a different question. It either be adopted to meet economic interests or be adopted purposefully for meeting sustainability goals, or a certainty that it will serve both. In Paper I, we can see such strong advocacy, that is generally rooted in serving both a social-environmental function and preserving business interests grounded in certainty (narrative 1 and narrative 2). This can act as a driver for drone adoption, can be a significant barrier for drones as they can undermine other systemic effects. Such optimism raise concerns in literature about the risk of technology-led transitions that fail to address deeper socio-

cultural drivers and in many ways reinforce existing unsustainable systems (Farla et al., 2012). It can lead policymakers to focus on new "technological fixes" while overlooking more fundamental sustainability issues such as rising transport demand and resource consumption that drive unsustainable mobility. These discourses determine how rationales are created for public investment to position drones as aligned with broader transport and climate agendas or not. It also shows conflicting beliefs around what constitutes "responsible innovation," with some actors advocating for rapid technological deployment and some prioritizing public legitimacy, safety, and environmental assessment.

Negative discourse, shows, creates, and reinforces distrust and disinterest in the technology and thus delays or can even fail to gain support from the regime. In Paper I, disinterest in drones is rooted within the negative aviation discourse, distrust in technology, or simply not prioritizing it (narrative 3). While narratives (like narrative 3) that are pessimistic of drones or technology in general act as a barrier for adoption, this pessimism can also be interpreted as dissolving or redistributing interest away from drones as a tool for purposive transitions, making them vulnerable to market interests to propel drone adoption efforts to meet user convenience, and economic interests, and not particularly long-term environmental goals. However, such pessimistic narratives are also emblematic of the discourses within sustainable futures that question the need for "technological fixes", questions their legitimacy in meeting sustainability goals and focuses on the structural and behavioural changes that enables change in mainstream ways of consumption (Feola & Jaworska, 2019).

Thus, these different narratives can contribute to a discursive environment for drones which can be interpreted as to have a duality that can act both as barriers and drivers while observing from current state of drone emergence. The outcome of which is influenced by many factors such as general change in attitudes, knowledge and experience with the technology enabled through technological developments, political interests, technological developments, environmental values, power and influence, or contextual impacts after use or adoption of the technology, which is to be determined with time.

6.2.2 Barriers for drone adoption

CO2 performance compared to E-vans

A key barrier for transition toward drone-based last-mile logistics is the rela-tively poor CO2 performance of large drones compared to E-vans. Paper IV shows E-trucks consistently exhibit substantially lower emissions across all use cases, ranging from 75% to 92% lower than drones per functional unit. This performance gap is mainly due to the high energy demand associated with drone take-off and landing, which becomes particularly inefficient in contexts where delivery distances are short, and the number of stops is high. The sensi-tivity analysis in Paper IV demonstrates that only under specific operational conditions, such as significantly shortened routes and major improvements in drone energy efficiency (which can only be achieved currently with smaller drones), could drones begin to approach performance of E-vans. Furthermore, in Paper V, small drone can support electric trucks in achieving higher energy

savings. As a result, the current emissions profile positions electric vans and trucks as the clear environmental choice, which means that large drone cannot be chosen over E-vans from an emissions and energy standpoint, which are supported by other assessments in the field (Kirschstein, 2021; Kumar et al., 2025). From this it also implies that E-trucks, that carry much higher payloads, would also be more efficient than large drones carrying out the same operation. This can act as a barrier in prioritising drones particularly large ones as an alternative in a climate where electric ground vehicle priority is both new and high, moreover, create competition for limited resources. However, it is important to approach this from a life-cycle perspective, as manufacturing of drones and E-vans may have high emissions, which has not been studied in this thesis.

Rebound effects

The uncertainties and risks of drone use (Filippone & Barakos, 2020) also stand as a barrier despite the ideas of its use in localized contexts or addressing immediate needs. The unintended system-wide consequences of a technology, or "rebound effects" pose long-term sustainability challenges (Binswanger, 2001). Rebound effects may arise from increased accessibility and convenience, potentially stimulating higher delivery demand, encouraging the decentralisation of warehouses, or driving the expansion of vertiport networks. These dynamics could contribute to land-use change and urban sprawl rather than a reduction in overall environmental pressure (Rendón et al., 2021; Straubinger, 2019). Moreover, by making fast, accessible, cheap, and high-volume delivery more attainable, they may increase consumption and ultimately enable unsustainable patterns and not resolve them. Such paradoxical externalities can act as a barrier to achieve sustainable transitions using drones. Risks and potential rebound effects present another important barrier to the purposive adoption of drones within sustainable transport transitions.

Lack of bottom-up frameworks

Across Paper I, II, and III evidence of the absence of bottom-up frameworks. From Paper I, current governance structures are seen to remain largely top down, with unclear division of responsibility and limited involvement of regional and municipal actors. This lack of distributed governance slows both coordination and learning processes, and risks reinforcing existing regime structures rather than supporting experimentation and niche development (Rip & Kemp, 1998) which can be a significant barrier to guiding and enabling purposive development of drones to contribute to sustainability transitions. While international and national regulation can create enabling conditions, sustainable adoption requires governance mechanisms that involve local authorities, planners, and communities in formulating specifics on why and how drones are integrated. For example, Paper II demonstrates that many of the key decisions required for drone integration such as vertiport placement, airspace zoning, charging infrastructure, public space use, requires bottom-up decisions. Issues such as noise, safety, privacy, land and airspace use require involvement of multi-level actors (national-regional-local) and context sensitive policy

development. Paper III adds a systems perspective by showing that the environmental and logistical performance of drone transport varies significantly depending on context. Integration with existing logistics infrastructure requires the involvement of transport actors, who also face regulatory barriers at their end (seen in Paper III) but are not involved to large extent (seen in Paper I) thus risking lack of user perspective. Bottom-up frameworks are essential as they provide agency to civil society and local actors to align technology with societal needs (Feola & Jaworska, 2019).

Public acceptance

Paper III identifies key public concerns including noise, privacy, safety, visual disturbance, environmental impacts, affordability, and property rights, and public hold expectations for regulations that address such challenges (EASA, 2021). Papers I and III highlight that such concerns must be addressed early in planning and implementation. An added concern is job losses as mentioned in Paper I, which can be argued that drones can create new jobs that require new skills. However, if decision-making involves choices that are based on the cost aspect alone, the high cost-effectiveness seen in Paper IV due to automation may also act as a social concern to lose jobs faster than can be created. Çetin et al. (2022) discusses how these factors can be addressed using mitigation factors implemented while conducting drone flights, albeit associated with resulting trade-offs. Public acceptance factors influence public attitudes, political decisions, and planning processes, particularly in relation to parts that will need to be addressed by technology policy (Grunwald, 2000). Thus, it can act as a barrier for drones that the industry and public authorities must deal with collectively.

Planning and coordination complexities

As the technology is still emerging, local authorities are unfamiliar with planning for new systems or infrastructure that drones necessitate. As Paper IV and V demonstrate, planners must identify where drones meaningfully improve logistics performance and design systems around those strengths rather than simply replacing existing modes. However, Papers II and III also reveal the complexity of cross-sector alignment. Sufficiency of local electricity grids capacities for introduction of electric vehicles (Palm, 2021; Pillai et al., 2012), especially when drones operate alongside other electrified freight systems such as electric trucks. These infrastructure demands imply that energy systems and logistics networks must evolve in coordinated ways. Drone integration involves coordination among aviation authorities, municipalities, logistics firms, telecommunications providers, emergency services, and energy operators. For each of these sectors, drones introduce new risks and planning requirements.

Paper III highlights that drone integration generates both synergies and trade-offs. Operational choices can amplify or reduce these effects (Barrado et al., 2020). For instance, lower altitude flight may improve energy efficiency but worsen noise and privacy impacts. Thus, integrating new planning practices with emerging technologies requires new forms of policy mixes that accommodate expertise, learning, and coordination (Park et al., 2024). At the

same time, avoiding radical disruption of existing practices, collaborating across sectors, and planning with inherent trade-offs all add to the complexity of planning (Kemp et al., 2007). Decision-makers need to balance operational efficiency with social and environmental impacts, while coordinating infrastructure, governance arrangements, public expectations, and business actors in policy.

Regulation has been considered one of the biggest barrier to drone deployment (Cohen et al., 2021) and Paper II in this thesis identifies the drone ecosystem and the aspects that require policy attention. From Paper I we can see gaps also due to fragmented discursive environment and responsibilities as one contributor to this barrier. Paper I shows how local actors expect private actors to lead and take up risks, while private actors look for public actors to address procedural issues. When public bodies lack interest to prioritise an innovation or assume it will happen through actors with business interests, a clear leadership role for drone development and critical dialogue around the technology in alignment with broader energy-transition and sustainability goals risk being sidelined. Such planning and coordination complexities can be a major barrier to overcome.

6.2.3 Drivers for drone adoption

CO2 performance compared to diesel vans

From a CO2 performance perspective, their lower emission particularly when compared with conventional diesel van may represent a potential driver for adoption within sustainability transitions. As discussed earlier, Paper IV indicates that large electric drones can exhibit lower CO2 emissions than diesel-fuelled IC-vans across the evaluated delivery scenarios, even when overall energy use remains higher. Given the context-dependent nature of these outcomes, drones can provide lower-carbon options in areas where electrified logistics are not yet widespread, thus positioning them as a potentially complementary technology within ongoing transport transitions. However, it is important to approach this from a life-cycle perspective, as manufacturing of drones may have high emissions, which has not been studied in this thesis. Additionally, if biofuels (like HVO seen in Paper IV) are being introduced instead of diesel in IC vans, then case based assessments need to be made before choice.

Flexibility and accessibility in low density areas

Flexibility and accessibility benefits can be seen in many ways throughout this thesis. Their ability to bypass road infrastructure, serve remote and low-density areas, and operate rapidly makes them interesting for emergency deliveries and rural logistics (Ahmed et al., 2020). Smaller drones that can be combined with existing logistics systems also adds to creating more flexibility in transport patterns during delivery logistics (Jazairy et al., 2024). Additionally, a new mode of local transport can provide new and flexible ways of envisioning future transport systems, thus driving interest in them.

Top-down policy frameworks

As seen in Paper II, top-down policy developments, particularly through the European Aviation Safety Agency are creating clearer regulatory pathways (EASA, 2022, 2023, 2024). Such EU-level developments create momentum where national transport bodies get involved and thus initiate alignment across national and regional governance structures, that could stimulate local actors to start planning for drone use. These policy signals act as drivers by reducing uncertainty and enabling investment. They also reinforce the perception that drones contribute to strategic goals such as decarbonisation, digitalisation, and competitiveness (EU, 2022b). This policy framing creates a permissive environment for experimentation and development of niche technology even if local actors remain uncertain.

Energy benefits of drone-assisted deliveries during use

Paper V demonstrates energy benefits of integrating drones with trucks in hybrid systems. These hybrid systems reduce unnecessary short truck trips, optimise routing, and reduce energy use per delivery. The energy reduction remains significant across multiple scenarios when combined with E-truck and IC-trucks. Their environmental impact does depend on the make-up of electricity from the grid (Lewis et al., 2021), and on their energy use for production and end-of-life phases (Figliozzi, 2020). However, it is useful to note that in the life-cycle of IC engine vehicles, the use-phase has highest energy consumption (Mayyas et al., 2017). Besides this, providing additional energy reducing capacity to electric truck systems is an added advantage. Thus, this result can be seen as a driver while opting to work with such systems.

Operational improvements

Cost and time are two of the operational improvement factors that can be seen from Paper IV. As mentioned previously in RQ1, the LCC results, and high speeds can provide major incentives for adopting drones to improve operational efficiency in transport logistics. It highlights the potential for drones to be immediately applicable for healthcare and other public services that require improved operations with the benefit of low costs. This also comes as an advantage for logistics companies that want to meet increased demand for speedy deliveries, provide improved and cost-efficient services for customers. Although it can be argued that economics or costs is an important dimension of sustainability, this argument can be seen as an adoption pressure that is expected and reinforced by the incumbent regime and macroeconomic landscape, which is how it is considered in MLP (Geels & Kemp, 2012). Therefore, while cost-effectiveness presents as a driver for technology adoption, environmental implications of such services need to be considered in light of earlier mentioned rebound effects and entrenched unsustainable patterns.

6.3 Drone implementation to meet local needs

RQ 3. How to address the emergence of drones to meet local sustainability needs?

Addressing the emergence of drones as a transport can be thought of as an immediate planning and deployment concern, and for long-term transition strategy for decarbonization and meeting local transport needs. From Research question 1 and 2, it becomes evident that drones bring forth potential that are adaptive in nature and could help in incremental improvements for sustainability transitions, and risks that can be detrimental. Drones can transform existing systems by bringing new complexities (see in the appended papers) that need to be managed proactively. Accordingly, Paper II presents a policy planning framework that is of relevance here. We can further arrive at the following general prerequisites that are needed to address these complexities.

Drone governance require policy making processes across local, regional, national, and supranational levels. National and European aviation authorities provide a strong top-down framework focusing on certification, airspace safety, compliance, etc while local and regional bodies need to tackle bottom-up priorities related to planning for local infrastructure (Paper II and III). Such multi-level set-up requires high coordination to deal with the high level of uncertainty that comes with dependence on the involvement of high number of actors (Liesbet & Gary, 2003). The lack of recognition of roles and responsibilities of actors in Paper I shows a lack of organization and co-ordination between different actors, which leads to governance gaps and uncertainty in implementation. Bottom-up governance becomes critical because many planning aspects and sustainability impacts of drones require expertise of local actors and their decision-making capacity to meet local needs. The policy framework developed in Paper II bridges this gap in multi-level governance model and emphasizes the need for coordination that allows local actors to influence implementation according to contextual needs.

Additionally, cross-sectoral coordination is also needed to coordinate the functionality and adoptability of drones as shown in Paper II. Cross-sectoral policymaking requires attention to both interactions within a single sector and that across multiple sectors at various levels of government, as these vertical mixes significantly increase policy complexity and demand greater coordination, integration, and adaptive governance to reduce conflicts, fragmentation, and implementation failures (Howlett & Del Rio, 2015).

As seen particularly in Paper IV, V and III, drone impacts and systems dynamics is highly context-dependent and varies by application, operating conditions, and systems design. Therefore, during this thesis, the perspective has shifted between what are the impacts of drones and what conditions allow drones to perform a certain way. The outcomes from RQ1 which shows how drone use can support sustainability transitions through hybridization towards phasing out fossil-use, also requires positioning drones within specific contexts where this can be achieved, as well as broader decarbonization strategies in

line with other transport technologies. This requires multi-phase transition management, and in ways that are deliberately designed to address different types of activities and actor roles across transition phases, guiding innovation processes through learning and coordination to move towards desired direction (Loorbach, 2010). Local pilot projects, living labs, and testbeds allow experimentation and feedback in controlled spaces (as seen in Paper II) which can enable this. In Sweden, pilot projects and co-creation have already begun in small niches in various parts to co-create pilot projects and living labs (Skellefteå, 2025). Introducing drones for decarbonization is thus, a multi-phase approach where drones are introduced first where they have clearer potential to outperform incumbent technologies, such as emergency deliveries, rural logistics, and hybrid truck-drone systems that this thesis indicates. Concepts such as air taxis, that are delimited in this thesis, are of higher radicality and risks, and require more learning for understanding their implications on existing systems prior to investing in technology and infrastructure (Ahmed et al., 2021).

Besides, local initiative-based learning, socio-technical assessment along with sustainability assessments should guide policy development (Geels, 2002). Socio-technical analysis helps critically-question the feasibility of transitions scenarios, and provides indications regarding choices, pathways and interventions in a phased manner (Turnheim et al., 2015). Choices for drone application can thus be guided by sustainability indicators, while carefully considering barriers, drivers, and trade-offs grounded in local contexts. Responsibility and leadership are needed to navigate through conflicting narratives, organize diverse stakeholders, foster collaboration, and ensure coordination across sectors. Creating safe spaces for co-creation before full-scale deployment is crucial, allowing actors to experiment, learn, and adapt while working with existing regulations, standards, and assessment methods. Early attention should also be given to circularity, systemic demand measures, and behaviour-oriented approaches, including opportunities to reuse or repurpose existing infrastructure (Volakakis & Mahmassani, 2024).

The following sub-section provides the implications of the findings of this thesis for practitioners that provides more insights into how drone emergence can be addressed.

6.3.1 Implications for Policymakers

The complex and cross-sectoral nature of drone emergence is evident in this thesis. Coordinated governance across local, regional, national, and EU levels is essential to provide clear policy direction and regulatory coherence. Cross-sectoral coordination is necessary to ensure managing the benefits and risks of implementing new technologies like drones. Strengthening of multi-level and cross-sectoral governance is considered necessary, along with making policy processes to be proactive, coordinated, and easily interpretable to varied actors. Setting the vision of drone development in line with policy mixes for sustainability goals is recommended (Howlett & Del Rio, 2015). Drone policies should explicitly require sustainability assessments and prioritise use cases that demonstrate emission reductions and energy savings, while avoiding

applications that conflict with climate goals. Impact assessments should be integrated at early stages using indicators such as energy use, greenhouse gas emissions, noise, biodiversity impacts, grid capacity, equity, land use change, and public acceptance to ensure alignment with local sustainability strategies. Additionally, such assessment tools that assess combinations of different types of vehicles also help make technological choices. Besides, policymakers must handle drone, and any alternative vehicle integration as a demand-side issue by regulating high-energy deliveries or using pricing mechanisms to restrict excessive demand.

While such action requires high external resources, it is also encouraged to focus on institutional learning and capacity building as managing such transitions requires internal capabilities. For example, regions and municipalities must develop competencies in considering the vertical dimension in urban and rural plans, and should be particularly involved in including airspace zoning, landing and take-off site identification, planning for vertiports, etc. Therefore, training programmes on emerging technology, interdisciplinary planning teams, and collaboration with academia and stakeholders can improve decision-making capacity and institutional practices.

Managing drone innovation is also a multiphase process (Barrado et al., 2020), that takes the technology and its infrastructure from immaturity to maturity, experimentation to implementation, varied levels of automation, and from goods delivery to more complex passenger transport. As seen in transition studies, the role of public institutions is therefore seen as interchangeable between "*facilitator–stimulator–controller–director*" depending on the ongoing phase (Kern & Howlett, 2009).

Policymakers should establish transparent communication platforms that share information on environmental and safety impacts, risks, mitigation measures, and trade-offs, while providing agency to communities and enabling their input to be integrated into decision-making process.

6.3.2 Implications for Industry

From the findings of the thesis regarding discursive lock-ins in (Paper I), it is encouraged for industry actors, particularly drone and aviation industry, to be engaged in and have discussions about technology and its broader systemic impacts (such as rebound effects, urban sprawl), sustainable futures, conflicting narratives, and various sustainability transitions perspectives. This is particularly pertinent to the drone and aviation industry, as this can encourage learning and understanding perspectives that lie beyond business interests. It would be beneficial in understanding the complexity of landscape developments, long-term consequences, perspectives of local bodies, immediate transport needs, and provide ways to be reflexive, and innovate in co-creative ways to contribute to sustainable transitions.

From Paper II, co-creation and transparency play a crucial role in strengthening the legitimacy and effectiveness of drone-related innovations. Collaboration with researchers, environmental experts, and local authorities can support shared learning and better alignment with local sustainability goals. Being transparent about interests, outcomes, and limitations improves credibility and

builds trust among actors and involved communities. Addressing public acceptance is also an important task for industry for responsible innovation. Public acceptance is regarded as essential for niche development and for establishing the legitimacy required for technological adoption (Grunwald, 2000). However, public acceptance should emphasize the development of drones that are acceptable to citizens, rather than attempting to model citizens' acceptance of drones (Dolata & Schwabe, 2023).

When considering drone-based delivery systems, designing systems that minimise take-off and landing energy, optimise battery efficiency, and support modular payloads are needed to improve energy performance. Hybrid systems can be used for energy reduction without requiring extensive new infrastructure, which might be helpful to improve delivery logistics. Drone system development should facilitate integration with existing logistics systems, particularly hybrid truck-drone operations to plan for complementing conventional delivery modes.

Drones can be critically explored for new business models such as reduce conventional aviation dependency for business at regional airports, bringing in alternative mobility solutions, encouraging on-site renewable energy infrastructure, and creating new practices which rely on innovating with existing infrastructure.

7. Future research avenues

This research aimed to explore civilian drones as a mode of transport within the sustainable transport transition context. At an early stage of drone emergence, this thesis provides foundational understanding on the potential effects that the emerging technology has on current transition pathways with interrelated technological, social, environmental, and policy factors. Particularly for contemplating use of emerging technologies for cross-sectoral transitions like energy transition in transportation such perspective is deemed critical to support decision-making processes that align with sustainability goals and local needs. This thesis thus opens crucial avenues for future research.

First, a limitation of this thesis is the lack of inclusion of infrastructure construction and maintenance, as well as the production and end-of life phases of the drone vehicle while developing comparative assessments with other modes of transport. It is recommended for future studies should incorporate this perspective as drones and its infrastructure are more mature, to make reliable assessments. Future studies are recommended to analyse trade-offs between system design parameters, environmental impacts, and societal acceptance over time. Additionally, modelling and evaluating energy use and emissions of multimodal drone-based logistics is an open area of research, for different combinations of vehicles, systems design, and operating conditions. Moreover, lack of impact studies situated in Swedish context highlights a key research need.

Future work should broaden the planning perspective to consider the reallocation and creative use of space released through modal shifts, a perspective that is essential for sustainable transport futures (Banister, 2008). Future studies are recommended to examine how drones could impact road-based traffic or be of service in areas that currently have issues with transport infrastructure.

Finally, this thesis highlights the need to examine critical narratives, systemic lock-ins, and trade-offs between technological potential and system-level effects, in order to better understand how emerging technologies may reinforce or challenge existing transport and energy regimes. Such engagement is essential not only for adapting existing systems, but for transforming them to achieve multifaceted sustainability and energy transition goals.

References

Afonso, F., Ferreira, A., Ribeiro, I., Lau, F., & Suleman, A. (2021). On the design of environmentally sustainable aircraft for urban air mobility. *Transportation Research Part D: Transport and Environment*, *91*. https://doi.org/10.1016/j.trd.2020.102688

Ahmed, S. S., Fountas, G., Eker, U., & Anastasopoulos, P. C. (2021). Are we willing to relocate with the future introduction of flying cars? An exploratory empirical analysis of public perceptions in the United States. *Transportmetrica A: Transport Science*, 1-28. https://doi.org/10.1080/23249935.2021.1916643

Ahmed, S. S., Hulme, K. F., Fountas, G., Eker, U., Benedyk, I. V., Still, S. E., & Anastasopoulos, P. C. (2020). The Flying Car—Challenges and Strategies Toward Future Adoption. *Frontiers in Built Environment*, *6*. https://doi.org/10.3389/fbuil.2020.00106

AIRMOUR. (2023). *D5.1 CO2 Life Cycle Assessment for Emergency Medical Service UAM Concepts*.

AIRMOUR. (2024). *Public Final Report* https://airmour.eu/public-final-report-deliverable-1-3/

Al Haddad, C., Chaniotakis, E., Straubinger, A., Plötner, K., & Antoniou, C. (2020). Factors affecting the adoption and use of urban air mobility. *Transportation Research Part A: Policy and Practice*, *132*, 696-712. https://doi.org/10.1016/j.tra.2019.12.020

Alvesson, M., & Sköldberg, K. (2000). *Reflexive Methodology: New Vistas for Qualitative Research*. Sage.

Aster, H., Zeilerbauer, L., & Lindorfer, J. (2025). Environmental Impacts of Drone Delivery: A Comparative Meta-Analysis and Standardised LCA Metrics. *Journal of Sustainability*, *1*(2). https://doi.org/10.55845/jos-2025-1249

Aydin, B. (2019). Public acceptance of drones: Knowledge, attitudes, and practice [Article]. *Technology in Society*, *59*, 14, Article 101180. https://doi.org/10.1016/j.techsoc.2019.101180

Baik, S. G., Willits, S., Scherer, S., Jiao, J., & Zhang, P. (2025). Evaluating the Energy Savings from Truck-drone Hybrid Last-mile Delivery in Arbitrary Cities. *Available at SSRN 5388230*.

Baldisseri, A., Siragusa, C., Seghezzi, A., Mangiaracina, R., & Tumino, A. (2022). Truck-based drone delivery system: An economic and environmental assessment. *Transportation Research Part D: Transport and Environment*, *107*, 103296. https://doi.org/https://doi.org/10.1016/j.trd.2022.103296

Banister, D. (2008). The sustainable mobility paradigm. *Transport Policy*, *15*(2), 73-80. https://doi.org/https://doi.org/10.1016/j.tranpol.2007.10.005

Barrado, C., Boyero, M., Brucculeri, L., Ferrara, G., Hately, A., Hullah, P., Martin-Marrero, D., Pastor, E., Rushton, A. P., & Volkert, A. (2020). U-Space Concept of Operations: A Key Enabler for Opening Airspace to Emerging Low-Altitude Operations. *Aerospace*, *7*(3). https://doi.org/10.3390/aerospace7030024

Bauranov, A., & Rakas, J. (2021). Designing airspace for urban air mobility: A review of concepts and approaches. *Progress in Aerospace Sciences*, *125*. https://doi.org/10.1016/j.paerosci.2021.100726

Benzies, K. M., Premji, S., Hayden, K. A., & Serrett, K. (2006). State-of-the-Evidence Reviews: Advantages and Challenges of Including Grey Literature. *Worldviews on Evidence-Based Nursing*, *3*(2), 55-61. https://doi.org/https://doi.org/10.1111/j.1741-6787.2006.00051.x

Bergen, N., & Labonté, R. (2020). "Everything Is Perfect, and We Have No Problems": Detecting and Limiting Social Desirability Bias in Qualitative Research. *Qualitative Health Research*, *30*(5), 783-792. https://doi.org/10.1177/1049732319889354

Berkhout, F., Smith, A., & Stirling, A. (2004). Socio-Technological Regimes and Transition Contexts. *System innovation and the transition to sustainability: theory, evidence and policy. Edward Elgar, Cheltenham*, 48-75. https://doi.org/10.4337/9781845423421.00013

Binswanger, M. (2001). Technological progress and sustainable development: what about the rebound effect? *Ecological Economics*, *36*(1), 119-132. https://doi.org/https://doi.org/10.1016/S0921-8009(00)00214-7

Bourdieu, P. (1977). *Outline of a Theory of Practice* (R. Nice, Trans.). Cambridge University Press. https://doi.org/DOI: 10.1017/CBO9780511812507

Brace, I. (2018). *Questionnaire design: How to plan, structure and write survey material for effective market research*. Kogan Page Publishers.

Bracken, L. J., A., B. H., & and Whitman, G. (2015). Transdisciplinary research: understanding the stakeholder perspective. *Journal of Environmental Planning and Management*, *58*(7), 1291-1308. https://doi.org/10.1080/09640568.2014.921596

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77-101. https://doi.org/10.1191/1478088706qp063oa

Brunelli, M., Ditta, C. C., & Postorino, M. N. (2023). New infrastructures for Urban Air Mobility systems: A systematic review on vertiport location and capacity. *Journal of Air Transport Management*, *112*, 102460. https://doi.org/https://doi.org/10.1016/j.jairtraman.2023.102460

Bruni, M. E., Khodaparasti, S., & Perboli, G. (2023). Energy Efficient UAV-Based Last-Mile Delivery: A Tactical-Operational Model With Shared Depots and Non-Linear Energy Consumption [Article]. *IEEE Access*, *11*, 18560-18570. https://doi.org/10.1109/access.2023.3247501

Bryman, A. (2016). *Social research methods*. Oxford university press.

Bulusu, V., Onat, E. B., Sengupta, R., Yedavalli, P., & Macfarlane, J. (2021). A Traffic Demand Analysis Method for Urban Air Mobility. *IEEE Transactions on Intelligent Transportation Systems*, *22*(9), 6039-6047. https://doi.org/10.1109/tits.2021.3052229

Buschmann, P., & Oels, A. (2019). The overlooked role of discourse in breaking carbon lock-in: The case of the German energy transition. *Wiley Interdisciplinary Reviews: Climate Change*, *10*(3), e574.

Çetin, E., Cano, A., Deransy, R., Tres, S., & Barrado, C. (2022). Implementing Mitigations for Improving Societal Acceptance of Urban Air Mobility. *Drones*, *6*(2), 28. https://www.mdpi.com/2504-446X/6/2/28

Christensen, C. M. (2015). *The innovator's dilemma: when new technologies cause great firms to fail*. Harvard Business Review Press.

Cohen, A. P., Shaheen, S. A., & Farrar, E. M. (2021). Urban Air Mobility: History, Ecosystem, Market Potential, and Challenges. *IEEE Transactions on Intelligent Transportation Systems*, *22*(9), 6074-6087. https://doi.org/10.1109/tits.2021.3082767

Collazos, J. S. G., Ardila, L. M. C., & Cardona, C. J. F. (2024). Energy transition in sustainable transport: concepts, policies, and methodologies. *Environmental Science and Pollution Research*, *31*(49), 58669-58686. https://doi.org/10.1007/s11356-024-34862-x

Cucurachi, S., van der Giesen, C., & Guinée, J. (2018). Ex-ante LCA of Emerging Technologies. *Procedia CIRP*, *69*, 463-468. https://doi.org/https://doi.org/10.1016/j.procir.2017.11.005

Czarniawska, B. (2000). The uses of narrative in organization research. *Rapport nr.: GRI reports*(2000).

Darvishpoor, S., Roshanian, J., Raissi, A., & Hassanalian, M. (2020). Configurations, flight mechanisms, and applications of unmanned aerial systems: A review. *Progress in Aerospace Sciences*, *121*, 100694. https://doi.org/https://doi.org/10.1016/j.paerosci.2020.100694

Denyer, D., & Tranfield, D. (2009). Producing a systematic review.

Dolata, M., & Schwabe, G. (2023). Moving beyond privacy and airspace safety: Guidelines for just drones in policing [Article]. *Government Information Quarterly*, *40*(4), 18, Article 101874. https://doi.org/10.1016/j.giq.2023.101874

Dukkanci, O., Kara, B., & Bektaş, T. (2020). Minimizing energy and cost in range-limited drone deliveries with speed optimization. *Transportation Research Part C: Emerging Technologies*. https://doi.org/10.2139/SSRN.3314556

Durach, C. F., Kembro, J., & Wieland, A. (2017). A new paradigm for systematic literature reviews in supply chain management. *Journal of Supply Chain Management*, *53*(4), 67-85.

EASA. (2020). *U-space: a quick overview*. https://www.easa.europa.eu/en/what-u-space

Easy Access Rules for U-space (Regulation (EU) 2021/664), (2021a). https://www.easa.europa.eu/en/document-library/easy-access-rules/easy-access-rules-u-space-regulation-eu-2021664

EASA. (2021b). *Study on the societal acceptance of Urban Air Mobility in Europe*. Retrieved from https://www.easa.europa.eu/downloads/127760/en

EASA. (2022). *Prototype Technical Specifications for the Design of VFR Vertiports for Operation with Manned VTOL-Capable Aircraft Certified in the Enhanced Category (PTS-VPT-DSN)*. Cologne, Germany Retrieved from https://www.easa.europa.eu/en/document-library/general-publications/prototype-technical-design-specifications-vertiports

EASA. (2023). *Innovative Air Mobility and Services*. https://www.easa.europa.eu/en/domains/drones-air-mobility/drones-air-mobility-landscape/basics-explained/innovative-air-mobility-and-services

Easy Access Rules for U-space, (2024). https://www.easa.europa.eu/en/document-library/easy-access-rules/easy-access-rules-u-space-regulation-eu-2021664

Eißfeldt, H. (2020). Sustainable Urban Air Mobility Supported with Participatory Noise Sensing. *Sustainability*, *12*(8). https://doi.org/10.3390/su12083320

Eißfeldt, H., & Biella, M. (2022). The public acceptance of drones – Challenges for advanced aerial mobility (AAM). *Transportation Research Procedia*, *66*, 80-88. https://doi.org/https://doi.org/10.1016/j.trpro.2022.12.009

Energimyndigheten. (2025). *Energiläget i siffror*. Retrieved from https://www.energimyndigheten.se/energisystem-och-analys/nulaget-i-energisystemet/energilaget/

EU. (2020). *COMMISSION DELEGATED REGULATION (EU) 2020/1058 of 27 April 2020 amending Delegated Regulation (EU) 2019/945 as regards the introduction of two new unmanned aircraft systems classes*. European Union Aviation Safety Agency Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32020R1058&from=EN

EU. (2021a). *COMMISSION IMPLEMENTING REGULATION (EU) 2021/664 of 22 April 2021 on a regulatory framework for the U-space*. European Union Aviation Safety Agency Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021R0664&from=EN

EU. (2021b). *COMMISSION IMPLEMENTING REGULATION (EU) 2021/665 of 22 April 2021 amending Implementing Regulation (EU) 2017/373 as regards requirements for providers of air traffic management/air navigation services and other air traffic management network functions in the U-space airspace designated in controlled airspace*. European Union Aviation Safety Agency Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021R0665&from=EN

EU. (2021c). *COMMISSION IMPLEMENTING REGULATION (EU) 2021/666 of 22 April 2021 amending Regulation (EU) No 923/2012 as regards requirements for manned aviation operating in U-space airspace*. European Union Aviation Safety Agency Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021R0666&from=EN

EU. (2022a). *COMMISSION IMPLEMENTING REGULATION (EU) 2022/425 of 14 March 2022 amending Implementing Regulation (EU) 2019/947 as regards postponing the transition dates for the use of certain unmanned aircraft systems in the 'open' category and the date of application for standard scenarios for operations executed in or beyond the visual line of sight*. European Union Aviation Safety Agency Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R0425&from=EN

EU. (2022b). *COMMUNICATION FROM THE COMMISSION TO THE EUROPEAN PARLIAMENT, THE COUNCIL, THE EUROPEAN ECONOMIC AND SOCIAL COMMITTEE AND THE COMMITTEE OF THE REGIONS - 'A Drone Strategy 2.0 for a Smart and Sustainable Unmanned Aircraft Eco-System in Europe'*. Brussels: EUROPEAN COMMISSION Retrieved from https://transport.ec.europa.eu/system/files/2022-11/COM_2022_652_drone_strategy_2.0.pdf

EU. (2025). *Delivering the European Green Deal*. https://commission.europa.eu/topics/climate-action/delivering-european-green-deal_en

Farla, J., Markard, J., Raven, R., & Coenen, L. (2012). Sustainability transitions in the making: A closer look at actors, strategies and resources. *Technological Forecasting and Social Change*, *79*(6), 991-998. https://doi.org/https://doi.org/10.1016/j.techfore.2012.02.001

Feola, G., & Jaworska, S. (2019). One transition, many transitions? A corpus-based study of societal sustainability transition discourses in four civil society's proposals. *Sustainability Science*, *14*(6), 1643-1656. https://doi.org/10.1007/s11625-018-0631-9

Figliozzi, M. (2017). Lifecycle Modeling and Assessment of Unmanned Aerial Vehicles (Drones) CO2e Emissions. *Transportation Research Part D-transport and Environment*, *57*, 251-261. https://doi.org/10.1016/J.TRD.2017.09.011

Figliozzi, M. A. (2020). Carbon emissions reductions in last mile and grocery deliveries utilizing air and ground autonomous vehicles [Article]. *Transportation Research Part D: Transport and Environment*, *85*, Article 102443. https://doi.org/10.1016/j.trd.2020.102443

Filippone, A., & Barakos, G. N. (2020). Rotorcraft systems for urban air mobility: A reality check. *The Aeronautical Journal*, *125*(1283), 3-21. https://doi.org/10.1017/aer.2020.52

Folke, C., Carpenter, S. R., Walker, B., Scheffer, M., Chapin, T., & Rockström, J. (2010). Resilience Thinking

Integrating Resilience, Adaptability and Transformability. *Ecology and Society*, *15*(4). http://www.jstor.org/stable/26268226

forskning.se. (2023). Drönare med hjärtstartare ofta på plats före ambulans. https://www.forskning.se/2023/11/23/dronare-med-hjartstartare-ofta-pa-plats-fore-ambulans/

Garg, V., Niranjan, S., Prybutok, V., Pohlen, T., & Gligor, D. (2023). Drones in last-mile delivery: A systematic review on Efficiency, Accessibility, and Sustainability. *Transportation Research Part D: Transport and Environment*, *123*, 103831. https://doi.org/https://doi.org/10.1016/j.trd.2023.103831

Garousi, V., Felderer, M., & Mäntylä, M. V. (2019). Guidelines for including grey literature and conducting multivocal literature reviews in software engineering. *Information and Software Technology*, *106*, 101-121. https://doi.org/https://doi.org/10.1016/j.infsof.2018.09.006

Garrow, L. A., German, B. J., & Leonard, C. E. (2021). Urban air mobility: A comprehensive review and comparative analysis with autonomous and electric ground transportation for informing future research. *Transportation Research Part C: Emerging Technologies*, *132*. https://doi.org/10.1016/j.trc.2021.103377

Geels, F. (2024). *The Multi-Level Perspective on Sustainability Transitions: Background, overview, and current research topics*. https://doi.org/10.33774/coe-2024-c15gb

Geels, F., & Kemp, R. (2012). The multi-level perspective as a new perspective for studying socio-technical transitions. *Automobility in transition? A socio-technical analysis of sustainable transport*, 49-79.

Geels, F. W. (2002). Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study. *Research Policy*, *31*(8), 1257-1274. https://doi.org/https://doi.org/10.1016/S0048-7333(02)00062-8

Geels, F. W. (2004). From sectoral systems of innovation to socio-technical systems: Insights about dynamics and change from sociology and institutional theory. *Research Policy*, *33*(6), 897-920. https://doi.org/https://doi.org/10.1016/j.respol.2004.01.015

Geels, F. W. (2005). The dynamics of transitions in socio-technical systems: A multi-level analysis of the transition pathway from horse-drawn carriages to automobiles (1860–1930). *Technology Analysis & Strategic Management*, *17*(4), 445-476. https://doi.org/10.1080/09537320500357319

Geels, F. W. (2010). Ontologies, socio-technical transitions (to sustainability), and the multi-level perspective. *Research Policy*, *39*(4), 495-510. https://doi.org/https://doi.org/10.1016/j.respol.2010.01.022

Geels, F. W. (2012). A socio-technical analysis of low-carbon transitions: introducing the multi-level perspective into transport studies. *Journal of Transport Geography*, *24*, 471-482. https://doi.org/https://doi.org/10.1016/j.jtrangeo.2012.01.021

Geels, F. W. (2014). Regime Resistance against Low-Carbon Transitions: Introducing Politics and Power into the Multi-Level Perspective. *Theory, Culture & Society*, *31*(5), 21-40. https://doi.org/10.1177/0263276414531627

Geels, F. W. (2019). Socio-technical transitions to sustainability: a review of criticisms and elaborations of the Multi-Level Perspective. *Current Opinion in Environmental Sustainability*, *39*, 187-201. https://doi.org/https://doi.org/10.1016/j.cosust.2019.06.009

Geels, F. W., Kern, F., Fuchs, G., Hinderer, N., Kungl, G., Mylan, J., Neukirch, M., & Wassermann, S. (2016). The enactment of socio-technical transition pathways: A reformulated typology and a comparative multi-level analysis of the German and UK low-carbon electricity transitions (1990–2014). *Research Policy*, *45*(4), 896-913. https://doi.org/https://doi.org/10.1016/j.respol.2016.01.015

Geels, F. W., Sovacool, B. K., Schwanen, T., & Sorrell, S. (2017a). The Socio-Technical Dynamics of Low-Carbon Transitions. *Joule*, *1*(3), 463-479. https://doi.org/https://doi.org/10.1016/j.joule.2017.09.018

Geels, F. W., Sovacool, B. K., Schwanen, T., & Sorrell, S. (2017b). Sociotechnical transitions for deep decarbonization. *Science*, *357*(6357), 1242-1244. https://doi.org/doi:10.1126/science.aao3760

Geller, E. S. (2003). Leadership to Overcome Resistance to Change. *Journal of Organizational Behavior Management*, *22*(3), 29-49. https://doi.org/10.1300/J075v22n03_04

Goodchild, A., & Toy, J. (2018). Delivery by drone: An evaluation of unmanned aerial vehicle technology in reducing CO 2 emissions in the delivery service industry. *Transportation Research Part D: Transport and Environment*. https://doi.org/10.1016/J.TRD.2017.02.017

Greenflyway. (2022). *Slutrapport Green Flyway Testarena 2019-11-20- 2022-09-30*. https://www.greenflyway.se/wp-content/uploads/2022/12/Sluttrapport-Green-Flyway-hovedprosjekt-2022-11-23-film.pdf

Grunwald, A. (2000). Technology Policy Between Long-Term Planning Requirements and Short-Ranged Acceptance Problems. New Challenges for Technology Assessment. In J. Grin & A. Grunwald (Eds.), *Vision Assessment: Shaping Technology in 21st Century Society: Towards a Repertoire for Technology Assessment* (pp. 99-147). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-59702-2_6

Hermwille, L. (2016). The role of narratives in socio-technical transitions—Fukushima and the energy regimes of Japan, Germany, and the United Kingdom. *Energy Research & Social Science*, *11*, 237-246. https://doi.org/https://doi.org/10.1016/j.erss.2015.11.001

Holden, J., & Goel, N. (2016). *Fast-forwarding to a future of on-demand urban air transportation*. UBER.

Hoogma, R., Kemp, R., Schot, J., & Truffer, B. (2005). *Experimenting for sustainable transport: the approach of strategic niche management*. Taylor & Francis.

Howlett, M., & Del Rio, P. (2015). The parameters of policy portfolios: Verticality and horizontality in design spaces and their consequences for policy mix formulation. *Environment and Planning C: Government and Policy*, *33*(5), 1233-1245.

IEA. (2025). *Total final consumption, World, 2023* https://www.iea.org/world/energy-mix

Jaller, M., Pineda, L., Ambrose, H., & Kendall, A. (2021). Empirical analysis of the role of incentives in zero-emission last-mile deliveries in California. *Journal of Cleaner Production*, *317*, 128353. https://doi.org/https://doi.org/10.1016/j.jclepro.2021.128353

Jasanoff, S. (2016). *The ethics of invention: Technology and the human future*. WW Norton & Company.

Jazairy, A., Persson, E., Brho, M., von Haartman, R., & Hilletofth, P. (2024). Drones in last-mile delivery: a systematic literature review from a logistics management perspective. *The International Journal of Logistics Management*, *ahead-of-print*(ahead-of-print). https://doi.org/10.1108/IJLM-04-2023-0149

Johansson, L.-G. (2021). *Empiricism and Philosophy of Physics* (Vol. 00434). Springer. https://research.ebsco.com/linkprocessor/plink?id=5fc5a6c3-9551-3074-a896-76366e37fe33

Kai, W., Jacquillat, A., & Vaze, V. (2022). Vertiport planning for urban aerial mobility: An adaptive discretization approach. *manufacturing & service operations management*, *24*(6), 3215-3235.

Kaplan, T. J. (1993). *The argumentative turn in policy analysis and planning.* (F. Frank & F. John, Eds.).

Kasliwal, A., Furbush, N. J., Gawron, J. H., McBride, J. R., Wallington, T. J., De Kleine, R. D., Kim, H. C., & Keoleian, G. A. (2019). Role of flying cars in sustainable mobility. *Nature Communications*, *10*(1), 1555. https://doi.org/10.1038/s41467-019-09426-0

Kellermann, R., Biehle, T., & Fischer, L. (2020). Drones for parcel and passenger transportation: A literature review. *Transportation Research Interdisciplinary Perspectives*, *4*. https://doi.org/10.1016/j.trip.2019.100088

Kellermann, R., Biehle, T., & Mostofi, H. (2023). Modelling public attitude towards drone delivery in Germany [Article]. *European Transport Research Review*, *15*(1), 14, Article 38. https://doi.org/10.1186/s12544-023-00606-0

Kemp, R., Loorbach, D., & Rotmans, J. (2007). Transition management as a model for managing processes of co-evolution towards sustainable development. *International Journal of Sustainable Development & World Ecology*, *14*(1), 78-91. https://doi.org/10.1080/13504500709469709

Kern, F., & Howlett, M. (2009). Implementing transition management as policy reforms: a case study of the Dutch energy sector. *Policy Sciences*, *42*(4), 391-408. https://doi.org/10.1007/s11077-009-9099-x

Kim, J. G., Kim, S., Choi, D., Park, J., & Kim, H. K. (2024). Conceptual Design of Floating Vertiport Anchored with Taut Mooring Lines [Article]. *International Journal of Steel Structures*, *24*(6), 1463-1475. https://doi.org/10.1007/s13296-024-00920-w

Kim, S., Kim, T., Suh, K., & Jeon, J. (2023). Energy and environmental performance of a passenger drone for an urban air mobility (UAM) policy with 3D spatial information in Seoul. *Journal of Cleaner Production*, *415*, 137683. https://doi.org/https://doi.org/10.1016/j.jclepro.2023.137683

Kirschstein, T. (2020). Comparison of energy demands of drone-based and ground-based parcel delivery services. *Transportation Research Part D: Transport and Environment*. https://doi.org/10.1016/j.trd.2019.102209

Kirschstein, T. (2021). Energy demand of parcel delivery services with a mixed fleet of electric vehicles. *Cleaner Engineering and Technology*, *5*, 100322. https://doi.org/https://doi.org/10.1016/j.clet.2021.100322

Kirschstein, T., & Meisel, F. (2015). GHG-emission models for assessing the eco-friendliness of road and rail freight transports. *Transportation Research Part B: Methodological*, *73*, 13-33. https://doi.org/ https://doi.org/10.1016/j.trb.2014.12.004

Kitchenham, B. (2004). Procedures for performing systematic reviews. *Keele, UK, Keele University*, *33*(2004), 1-26.

Komasová, S., Tesař, J., & Soukup, P. (2020). Perception of drone related risks in Czech society. *Technology in Society*, *61*, 101252.

Kommun, T. (2022). *Förstudie -Hemleverans från landsbygdsbutiker med drönare*. https://torsby.se/download/18.1a4c845618383cccd5d5140e/1666070694933/Slutrapport%20fo%CC%88rstudie%20-Hemleverans%20fra%CC%8An%20landsbygdsbutiker%20med%20dro%CC%88nare.pdf

Kommun, T. (2024). *Pilotprojekt hemleverans från landsbygdsbutiker med drönare- Slutrapport*. https://torsby.se/download/18.671aed341932465297e8d88/1732471767186/Slutrapport%20Pilotprojekt%20-%20Hemleverans%20fra%CC%8An%20landsbygdsbutiker%20med%20dro%CC%88nare.pdf

Kremer, P., Haruna, F., Tuffour Sarpong, R., Agamah, D., Billy, J., Osei-Kwakye, K., Aidoo, P., Dodoo, D., & Okoh-Owusu, M. (2023). An impact assessment of the use of aerial logistics to improve access to vaccines in the Western-North Region of Ghana. *Vaccine*, *41*(36), 5245-5252. https://doi.org/https://doi.org/10.1016/j.vaccine.2023.06.036

Kumar, A., Prybutok, V., & Sangana, V. K. (2025). Environmental Implications of Drone-Based Delivery Systems: A Structured Literature Review. *Clean Technologies*, *7*(1).

Kvale, S., & Brinkmann, S. (2009). *InterViews: Learning the Craft of Qualitative Research Interviewing.* (2nd Edition ed.). Sage.

Lang, D. J., Wiek, A., Bergmann, M., Stauffacher, M., Martens, P., Moll, P., Swilling, M., & Thomas, C. J. (2012). Transdisciplinary research in sustainability science: practice, principles, and challenges. *Sustainability Science*, *7*, 25-43.

Legere, M. M. (2019). Meta-Analysis of Public Acceptance of Unmanned Aircraft Systems in the United States.

Levin, K., Cashore, B., Bernstein, S., & Auld, G. (2012). Overcoming the tragedy of super wicked problems: constraining our future selves to ameliorate global climate change. *Policy Sciences*, *45*(2), 123-152. https://doi.org/10.1007/s11077-012-9151-0

Lewis, E., Ponnock, J., Cherqaoui, Q., Holmdahl, S., Johnson, Y., Wong, A., & Oliver Gao, H. (2021). Architecting urban air mobility airport shuttling systems with case studies: Atlanta, Los Angeles, and Dallas. *Transportation Research Part A: Policy and Practice*, *150*, 423-444. https://doi.org/10.1016/j.tra.2021.06.026

LFV. (2023). *Översyn av det undre luftrummet redovisning av regeringsuppdrag*. Retrieved from https://www.regeringen.se/contentassets/61e1856b68db4f57ab40eb78ddbd4567/oversyn-av-det-undre-luftrummet.pdf

Liesbet, H., & Gary, M. (2003). Unraveling the Central State, but How? Types of Multi-level Governance. *American Political Science Review*, *97*(2), 233-243. https://doi.org/10.1017/S0003055403000649

Liu, Y., Lyu, C., Bai, F., Parishwad, O., & Li, Y. (2024). The role of intelligent technology in the development of urban air mobility systems: A technical perspective. *Fundamental Research*, *4*(5), 1017-1024. https://doi.org/https://doi.org/10.1016/j.fmre.2023.08.006

Loorbach, D. (2010). Transition management for sustainable development: a prescriptive, complexity-based governance framework. *Governance*, *23*(1), 161-183.

Loorbach, D. A. (2022). Designing radical transitions: a plea for a new governance culture to empower deep transformative change. *City, Territory and Architecture*, *9*(1), 30. https://doi.org/10.1186/s40410-022-00176-z

Lundvall, B.-A. (1992). *National systems of innovation: towards a theory of innovation and interactive learning* (Vol. 242). London Pinter.
Markard, J., Raven, R., & Truffer, B. (2012). Sustainability transitions: An emerging field of research and its prospects. *Research Policy*, *41*(6), 955-967. https://doi.org/https://doi.org/10.1016/j.respol.2012.02.013
Mavraj, G., Eltgen, J., Fraske, T., Swaid, M., Berling, J., Röntgen, O., Fu, Y., & Schulz, D. (2022). A Systematic Review of Ground-Based Infrastructure for the Innovative Urban Air Mobility. *Transactions on Aerospace Research*, *2022*(4), 1-17. https://doi.org/10.2478/tar-2022-0019
Mayyas, A., Omar, M., Hayajneh, M., & Mayyas, A. R. (2017). Vehicle's lightweight design vs. electrification from life cycle assessment perspective. *Journal of Cleaner Production*, *167*, 687-701. https://doi.org/https://doi.org/10.1016/j.jclepro.2017.08.145
Mohamed, A., & Mohamed, M. (2025). Unmanned Aerial Vehicles in Last-Mile Parcel Delivery: A State-of-the-Art Review. *Drones*, *9*(6).
Nordensvard, J., & Ketola, M. (2022). Populism as an act of storytelling: analyzing the climate change narratives of Donald Trump and Greta Thunberg as populist truth-tellers. *Environmental Politics*, *31*(5), 861-882. https://doi.org/10.1080/09644016.2021.1996818
Ochs, E. (1997). Narrative. In *Discourse as structure and process: Discourse studies: A multidisciplinary introduction* (pp. 185-207). SAGE Publications Ltd.
Oksman, V., & Kulju, M. (2022). Nordic study on public acceptance of autonomous drones. International Conference on Human-Computer Interaction,
Palm, J. (2021). Exploring Limited Capacity in the Grid: Actors, Problems, and Solutions [Original Research]. *Frontiers in Energy Research*, *9*. https://doi.org/10.3389/fenrg.2021.663769
Paredis, E. (2011). Sustainability Transitions and the Nature of Technology. *Foundations of Science*, *16*(2), 195-225. https://doi.org/10.1007/s10699-010-9197-4
Park, J., Kim, S., & Suh, K. (2018). A Comparative Analysis of the Environmental Benefits of Drone-Based Delivery Services in Urban and Rural Areas. *Sustainability*, *10*(3), 888. https://www.mdpi.com/2071-1050/10/3/888
Park, S. W. (2021). *Social acceptability of urban air mobility by aircraft category and autonomous phases* KDI School].
Park, T., Ki, J., & Ho, J.-Y. (2024). Sectoral innovation policy adopting socio-technical systems (STS) perspectives: The case of autonomous vehicles (AVs) in five leading countries. *Technological Forecasting and Social Change*, *209*, 123758. https://doi.org/https://doi.org/10.1016/j.techfore.2024.123758
Phillips, N., Lawrence, T. B., & Hardy, C. (2004). Discourse and Institutions. *The Academy of Management Review*, *29*(4), 635-652. https://doi.org/10.2307/20159075
Pierson, P. (2000). Increasing Returns, Path Dependence, and the Study of Politics. *The American Political Science Review*, *94*(2), 251-267. https://doi.org/10.2307/2586011

Pillai, J. R., Thøgersen, P., Møller, J., & Bak-Jensen, B. (2012). Integration of electric vehicles in low voltage danish distribution grids. 2012 IEEE power and energy society general meeting,

Qu, W., Xu, C., Tan, X., Tang, A., He, H., & Liao, X. (2023). Preliminary Concept of Urban Air Mobility Traffic Rules. *Drones*, *7*(1), 54. https://www.mdpi.com/2504-446X/7/1/54

Rahman, B., Bridgelall, R., Habib, M. F., & Motuba, D. (2023). Integrating Urban Air Mobility into a Public Transit System: A GIS-Based Approach to Identify Candidate Locations for Vertiports. *Vehicles*, *5*(4), 1803-1817. https://www.mdpi.com/2624-8921/5/4/97

Ramos, T. R. P., & Vigo, D. (2023). A new hybrid distribution paradigm: Integrating drones in medicines delivery [Article]. *Expert Systems with Applications*, *234*, 24, Article 120992. https://doi.org/10.1016/j.eswa.2023.120992

Ranhagen, U. (2020). *Co-Creation in Urban Station Communities: Summary of three lectures with a presentation of findings from the Project 2017-2019*. Chalmers tekniska högskola.

Ratei, P., Naeem, N., Prakasha, P. S., & Nagel, B. (2024). Sensitivity analysis of urban air mobility aircraft design and operations including battery charging and swapping [Article]. *CEAS Aeronautical Journal*, *15*(4), 1-15. https://doi.org/10.1007/s13272-024-00725-x

Raven, R. (2007). Niche accumulation and hybridisation strategies in transition processes towards a sustainable energy system: An assessment of differences and pitfalls. *Energy Policy*, *35*(4), 2390-2400. https://doi.org/https://doi.org/10.1016/j.enpol.2006.09.003

Rendón, M. A., Sánchez R, C. D., Gallo M, J., & Anzai, A. H. (2021). Aircraft Hybrid-Electric Propulsion: Development Trends, Challenges and Opportunities. *Journal of Control, Automation and Electrical Systems*, *32*(5), 1244-1268. https://doi.org/10.1007/s40313-021-00740-x

Richardson, B. C. (1999). Toward a Policy on a Sustainable Transportation System. *Transportation Research Record*, *1670*(1), 27-34. https://doi.org/10.3141/1670-05

Rimjha, M., Hotle, S., Trani, A., & Hinze, N. (2021). Commuter demand estimation and feasibility assessment for Urban Air Mobility in Northern California. *Transportation Research Part A: Policy and Practice*, *148*, 506-524. https://doi.org/10.1016/j.tra.2021.03.020

Rimjha, M., & Trani, A. (2021, 19-23 April 2021). Urban Air Mobility: Factors Affecting Vertiport Capacity. 2021 Integrated Communications Navigation and Surveillance Conference (ICNS),

Rip, A., & Kemp, R. (1998). Technological change. In *Human choice and climate change: Vol. II, resources and technology* (pp. 327-399). Battelle Press.

Roberts, C., & Geels, F. W. (2019). Conditions for politically accelerated transitions: Historical institutionalism, the multi-level perspective, and two historical case studies in transport and agriculture. *Technological Forecasting and Social Change*, *140*, 221-240. https://doi.org/https://doi.org/10.1016/j.techfore.2018.11.019

Rodrigues, T. A., Patrikar, J., Oliveira, N. L., Matthews, H. S., Scherer, S., & Samaras, C. (2022). Drone flight data reveal energy and greenhouse gas emissions savings for very small package delivery. *Patterns*, *3*(8). https://doi.org/10.1016/j.patter.2022.100569

Rothfeld, R., Fu, M., Balać, M., & Antoniou, C. (2021). Potential Urban Air Mobility Travel Time Savings: An Exploratory Analysis of Munich, Paris, and San Francisco. *Sustainability*, *13*(4). https://doi.org/10.3390/su13042217

Rotmans, J., Kemp, R., & Van Asselt, M. (2001). More evolution than revolution: transition management in public policy. *foresight*, *3*(1), 15-31.

Rotmans, J., & Loorbach, D. (2009). Complexity and Transition Management. *Journal of Industrial Ecology*, *13*(2), 184-196. https://doi.org/https://doi.org/10.1111/j.1530-9290.2009.00116.x

SAAB. (2024). *The drone that combats climate change: a collaboration between Sweden and Colombia*. https://www.saab.com/newsroom/press-releases/2024/the-drone-that-combats-climate-change-a-collaboration-between-sweden-and-colombia

Sälstrand, J. (2022). Drönarleverans sätter Tjockö på Sverigekartan. *Skärgården*. https://www.skargarden.se/dronarleverans-satter-tjocko-pa-sverigekartan/

Schaarschmidt, M., Bertram, M., & Knobloch, M. (2021). *Last mile drone delivery services: Adoption barriers before and during the COVID-19 pandemic*.

Scholz, R. W. (2017). The normative dimension in transdisciplinarity, transition management, and transformation sciences: New roles of science and universities in sustainable transitioning. *Sustainability*, *9*(6), 991.

SESAR. (2023). https://www.sesarju.eu/projects/CORUSXUAM

SESARJU. (2024). *Large scale demonstrations project. CORUS-XUAM - Concept of Operations for European U-space Services - Extension for Urban Air Mobility*. https://www.sesarju.eu/projects/CORUSXUAM

Sivakumar, M., & Tyj, N. M. (2021). A literature survey of unmanned aerial vehicle usage for civil applications. *Journal of Aerospace Technology and Management*, *13*, e4021. https://doi.org/https://doi.org/10.1590/jatm.v13.1233

Skellefteå, S. C. (2025). *Skellefteå Droneport*. https://www.skellefteasciencecity.se/en/projekt/skelleftea-droneport/

Smith, A. (2007). Translating Sustainabilities between Green Niches and Socio-Technical Regimes. *Technology Analysis & Strategic Management*, *19*, 427 - 450.

Smith, A., Dickinson, J. E., Marsden, G., Cherrett, T., Oakey, A., & Grote, M. (2022). Public acceptance of the use of drones for logistics: The state of play and moving towards more informed debate. *Technology in Society*, *68*. https://doi.org/10.1016/j.techsoc.2022.101883

Smith, A., Stirling, A., & Berkhout, F. (2005). The governance of sustainable socio-technical transitions. *Research Policy*, *34*(10), 1491-1510.

Sönmez, M., Pelin, C., Georgescu, M., Pelin, G., Stelescu, M., Nituica, M., Stoian, G., Alexandrescu, L., & Gurau, D. (2022). *Unmanned Aerial Vehicles – Classification, Types of Composite Materials Used in Their Structure and Applications*. https://doi.org/10.24264/icams-2022.I.11

Spangenberg, J. H. (2011). Sustainability science: a review, an analysis and some empirical lessons. *Environmental Conservation*, *38*(3), 275-287. https://doi.org/10.1017/S0376892911000270

Stebbins, R. (2001). *Exploratory Research in the Social Sciences*. SAGE Publications, Inc. https://doi.org/10.4135/9781412984249

Stolaroff, J., Samaras, C., O'Neill, E. R., Lubers, A., Mitchell, A. S., & Ceperley, D. (2018). Energy use and life cycle greenhouse gas emissions of drones for commercial package delivery. *Nature Communications*, *9*, null-null. https://doi.org/10.1038/s41467-017-02411-5

Straubinger, A. (2019). Policies addressing possible urban air mobility market distortions – a first discussion. *Transportation Research Procedia 41*, 64–66. https://doi.org/Straubinger, Anna

Sunil, E., Hoekstra, J., Ellerbroek, J., Bussink, F., Nieuwenhuisen, D., Vidosavljevic, A., & Kern, S. (2015). Metropolis: Relating airspace structure and capacity for extreme traffic densities. ATM seminar 2015, 11th USA/EUROPE Air Traffic Management R&D Seminar, FAA & Eurocontrol, Lisboa, Portugal.

Sunil, E., Hoekstra, J., Ellerbroek, J., Bussink, F., Vidosavljevic, A., Delahaye, D., & Nieuwenhuisen, D. (2016). How Do Layered Airspace Design Parameters Affect Airspace Capacity and Safety? The Influence of Traffic Structure on Airspace Capacity. ICRAT 2016, 7th International Conference on Research in Air Transportation, Philadelphia, USA.

SVT. (2022). *Här kommer drönare med leverans till Northvolt*. https://www.svt.se/nyheter/lokalt/vasterbotten/har-kommer-dronare-med-leverans-till-northvolt

Syed, N., Rye, M., Ade, M., Trani, A., Hinze, N., Swingle, H., Smith, J. C., Dollyhigh, S., & Marien, T. (2017). Preliminary considerations for odm air traffic management based on analysis of commuter passenger demand and travel patterns for the silicon valley region of California. 17th AIAA Aviation Technology, Integration, and Operations Conference,

Tan, L. K. L., Lim, B. C., Park, G., Low, K. H., & Yeo, V. C. S. (2021). Public acceptance of drone applications in a highly urbanized environment [Article]. *Technology in Society*, *64*, 14, Article 101462. https://doi.org/10.1016/j.techsoc.2020.101462

Taye, A., Wei, P., Pradeep, P., Jones, J., Bonin, T., & Eberle, D. (2024). Energy Demand Analysis for eVTOL Charging Stations in Urban Air Mobility. AIAA Aviation Forum and ASCEND, 2024,

Taylor, M., Saldanli, A., & Park, A. (2020, 8-10 Sept. 2020). Design of a Vertiport Design Tool. 2020 Integrated Communications Navigation and Surveillance Conference (ICNS),

Thelen, K. (1999). HISTORICAL INSTITUTIONALISM IN COMPARATIVE POLITICS. *Annual Review of Political Science*, *2*(Volume 2, 1999), 369-404. https://doi.org/https://doi.org/10.1146/annurev.polisci.2.1.369

Torraco, R. J. (2016). Writing Integrative Literature Reviews:Using the Past and Present to Explore the Future. *Human Resource Development Review*, *15*(4), 404-428. https://doi.org/10.1177/1534484316671606

Townsend, A., Jiya, I. N., Martinson, C., Bessarabov, D., & Gouws, R. (2020). A comprehensive review of energy sources for unmanned aerial vehicles, their shortfalls and opportunities for improvements. *Heliyon*, *6*(11). https://doi.org/10.1016/j.heliyon.2020.e05285

Trafikverket. (2025). *Fossilfritt transportsystem 2045*. https://bransch.trafikverket.se/for-dig-i-branschen/forskning-och-innovation/aktuell-forskning/

Transportstyrelsen. (2025). https://www.transportstyrelsen.se/sv/luftfart/flygplatser-flygtrafiktjanst-och-luftrum/luftrum/u-space/

Turnheim, B., Berkhout, F., Geels, F., Hof, A., McMeekin, A., Nykvist, B., & van Vuuren, D. (2015). Evaluating sustainability transitions pathways: Bridging analytical approaches to address governance challenges. *Global Environmental Change*, *35*, 239-253. https://doi.org/https://doi.org/10.1016/j.gloenvcha.2015.08.010

UNICEF. (2020). *How Drones Can Be Used to combat COVID-19*. Retrieved from https://www.unicef.org/supply/media/5286/file/%20Rapid-guidance-how-can-drones-help-in-Covid-19-response.pdf.pdf

Vattenfall. (2021). *Drönarleveranser kan ge klimatsmarta transporter*. https://www.vattenfall.se/fokus/trender-och-innovation/dronare/

VGRFokus. (2021). *Akutläkaren räddade livet vid hjärtstopp − med hjälp av drönare*. https://vgrfokus.se/2021/12/akutlakaren-raddade-livet-vid-hjartstopp-%E2%88%92-med-hjalp-av-dronare/

Volakakis, V., & Mahmassani, H. S. (2024). Vertiport Infrastructure Location Optimization for Equitable Access to Urban Air Mobility [Article]. *Infrastructures*, *9*(12), Article 239. https://doi.org/10.3390/infrastructures9120239

Walker, B., Holling, C. S., Carpenter, S. R., & Kinzig, A. (2004). Resilience, adaptability and transformability in social–ecological systems. *Ecology and Society*, *9*(2).

Wang, Y., Yang, S., Wang, X. V., & Wang, L. (2024). Research on truck-drone collaborative route planning for rural logistics delivery services. *Scientific Reports*, *14*(1), 31815. https://doi.org/10.1038/s41598-024-83149-1

Wells, P., Abouarghoub, W., Pettit, S., & Beresford, A. (2020). A socio-technical transitions perspective for assessing future sustainability following the COVID-19 pandemic. *Sustainability: Science, Practice and Policy*, *16*(1), 29-36. https://doi.org/10.1080/15487733.2020.1763002

Winter, S. R., Rice, S., & Lamb, T. L. (2020). A prediction model of Consumer's willingness to fly in autonomous air taxis. *Journal of Air Transport Management*, *89*, 101926. https://doi.org/https://doi.org/10.1016/j.jairtraman.2020.101926

Woodward, D. G. (1997). Life cycle costing—Theory, information acquisition and application. *International Journal of Project Management*, *15*(6), 335-344. https://doi.org/https://doi.org/10.1016/S0263-7863(96)00089-0

Yan, Y., Wang, K., & Qu, X. (2024). Urban air mobility (UAM) and ground transportation integration: A survey [Article]. *Frontiers of Engineering Management*, *11*(4), 734-758. https://doi.org/10.1007/s42524-024-0298-0

Yedavalli, P., & Mooberry, J. (2022). *An Assessment of Public Perception of Urban Air Mobility (UAM)*. Airbus. https://www.airbus.com/sites/g/files/jlcbta136/files/2022-07/Airbus-UTM-public-perception-study%20-urban-air-mobility.pdf

Yin, R. K. (2009). *Case study research: Design and methods* (Vol. 5). sage.

Yowtak, K., Imiola, J., Andrews, M., Cardillo, K., & Skerlos, S. (2020). Comparative life cycle assessment of unmanned aerial vehicles, internal combustion engine vehicles and battery electric vehicles for grocery delivery. *Procedia CIRP*, *90*, 244-250. https://doi.org/https://doi.org/10.1016/j.procir.2020.02.003

Papers
Associated papers have been removed in the electronic version of this thesis.
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